

Presentation

Mohamed Ali Abuella
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كلية التقنية الصناعية - مصراته
The College Of Industrial Technology_Misurata



SIU
CARBONDALE


UNC CHARLOTTE
Energy Production and Infrastructure Center (EPIC)



**HALMSTAD
UNIVERSITY**
**Northumbria
University**
NEWCASTLE



Presentation Outline

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graph TD; A[Presentation Outline] --> B[Introduction]; A --> C[Motivation]; A --> D[Research];
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Introduction

Motivation

Research

Introduction

Mohamed Abuella

<https://mohamedabuella.github.io>

<https://www.linkedin.com/in/mohamed-abuella/>

About Me..

An electrical engineer by training, traditionally is interested in Mathematical and Computational Analysis, Modeling and Optimization, and who is recently passionate in Artificial Intelligence and Data-driven Analytics.

A researcher works to modernize the electric grid and optimize its integration of distributed energy resources by applying descriptive, predictive and prescriptive analytics.

His broader interest involves utilizing Artificial Intelligence to foster Sustainability.

An adaptative to work in a diverse environment for an interdisciplinary research.

Introduction

<https://mohamedabuella.github.io/cv>

To sum it up in a broad sense, let's imagine that.. If my professional development was a book, its title would be **"Energy Systems Modeling and Analysis: Operation, Planning, and Integration."**

Thus, the chapters of this book would be as follows:

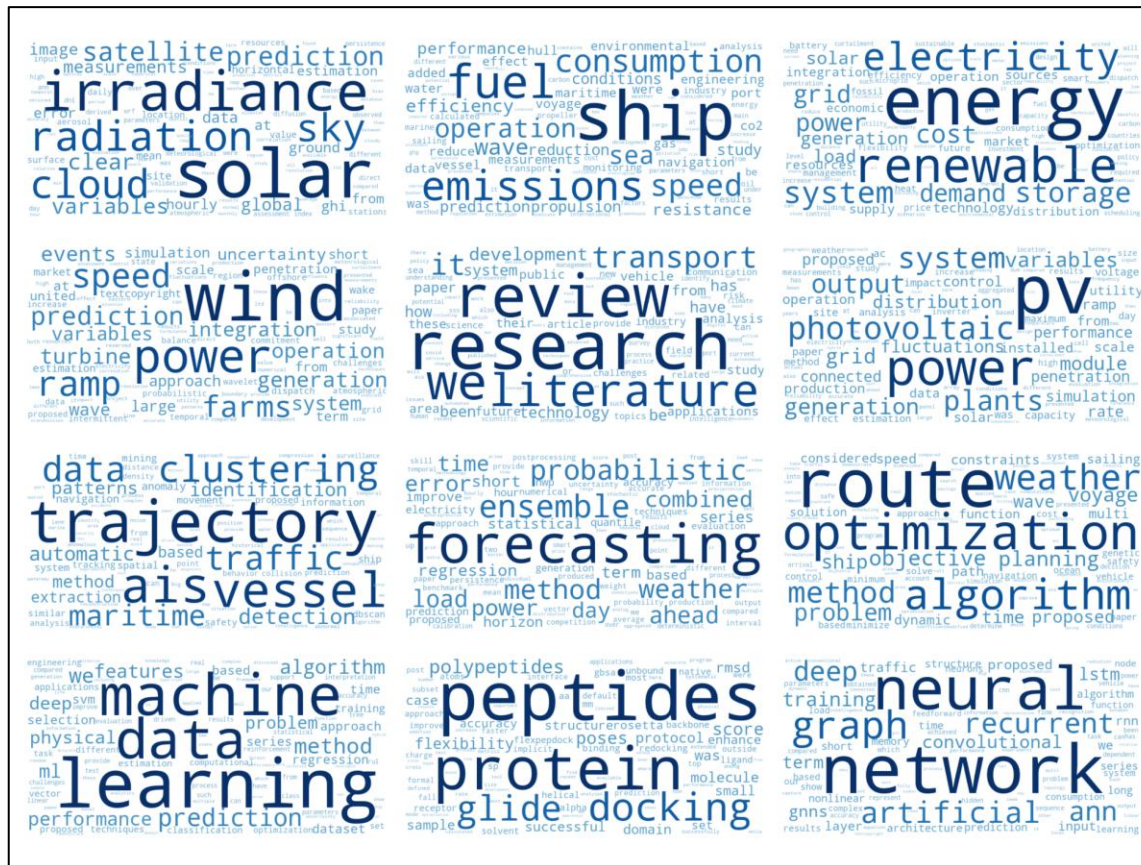
- Ch.1** Fundamentals of Electrical Engineering. This chapter covers Instrumentation & Control, Basics of Power Electronics such as Diodes & Thyristors as rectifiers, (maneuvered by applying Laws of Physics). With getting hands-on electrical installation & wiring and maintenance of electrical control equipment at pumping stations.
- Ch.2** Power Systems Analysis. It includes Power Flow and Faults Calculations, (applying Numerical Analysis methods, such as Newton methods, Differential eqs & Integrals, etc). Get hands-on some simulations of power systems and programmable logic controllers (PLC).
- Ch.3** Optimal Power Flow (OPF) and Security-Constrained Economic Dispatch (SCED). It is considering renewables as well, specifically for wind energy resources at the transmission level, (applying Optimization techniques). Get hands-on more of modeling and analysis of power systems.
- Ch.4** Optimize the Integration of Renewables into the Grid. Solar Power Modeling and Forecasting, (applying Descriptive, Predictive and Prescriptive Analytics, AI and ML techniques). Get hands-on data-driven analytics and become more familiar with conducting & publishing research.
- Ch.5** Postdoctoral Researcher at the Center for Applied Intelligent Systems Research (CAISR) at Halmstad University, (Integrated Academy-Industry Collaboration, Applying AI techniques). Dig into research on AI for Sustainability.
- Ch.6** Research Fellow at Northumbria University in Hydrogen Integration for Energy Transitions.

Introduction

In a nutshell, what I am often doing is finding the optimal & root values and curve fitting of nonlinear equations.

..But usually it is not as simple as that!

For more details, you may have a look at pdf copies of my [CV](#) and [Cloud of Key Skills & Interests](#).



Introduction

What can I bring to the team..?

- Transfer knowledge and skills to integrate the collaboration between academy and industry:
 - ✓ Energy Systems Modeling & Analysis, Optimization, AI, Geospatial Analysis.
 - ✓ Including Transportation and Maritime Systems.
 - ✓ Including Water and Wastewater experience
- An adaptatively to diverse environment and flexibility to multidisciplinary research:
 - ✓ Study and work in Middle East (Libya): Engineering, Electricity, Water and Wastewater.
 - ✓ Study and work in North America (USA): Energy, Optimization, Statistics, AI.
 - ✓ Work in Europe (Sweden): Energy, Transportation, Maritime, AI.
 - ✓ Work in Europe (United Kingdom): Hydrogen Integration for Energy Transitions.

Motivation

- Professional Advancement
 - ✓ Get an opportunity to collaborate and work with the experts of the field.
 - ✓ To transfer, improve, and acquire knowledge and skills.
- Personal Advancement
 - ✓ Better alignment with personal values and interests.
 - ✓ Better self-esteem.
 - ✓ Better financial security.

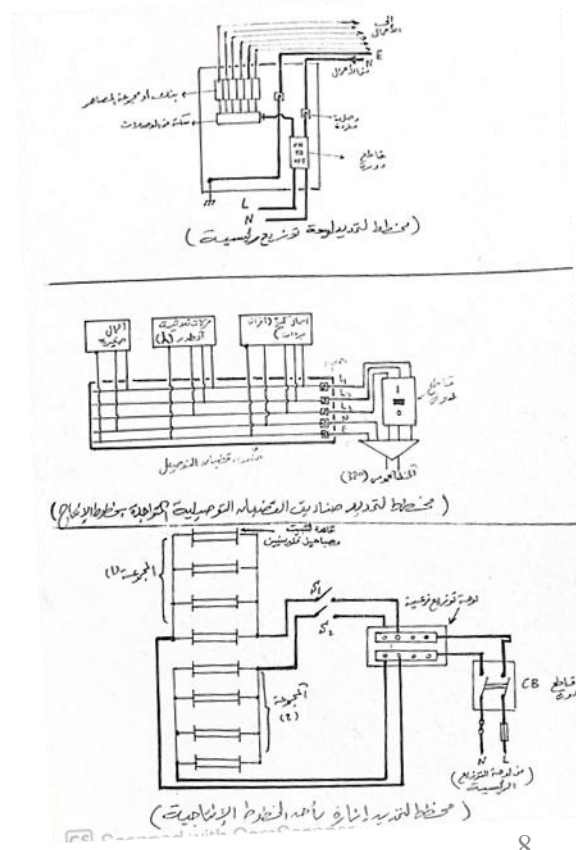
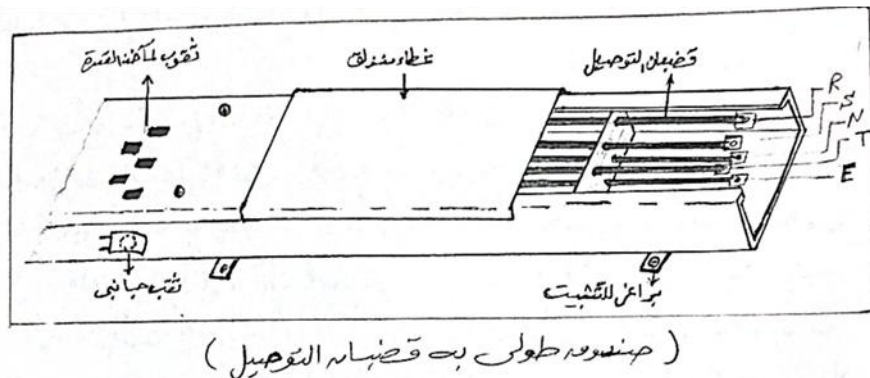
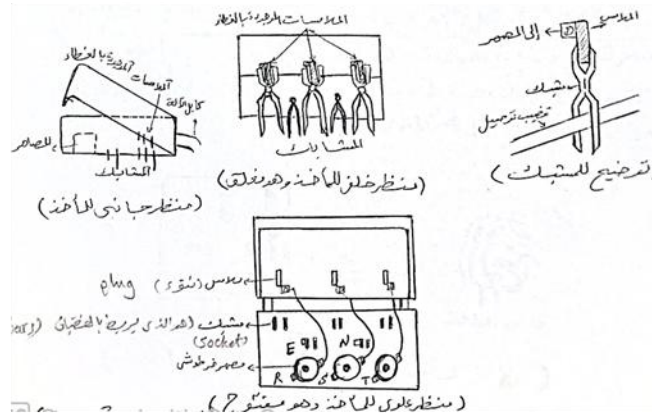
Research

Technical Report of Operation & Maintenance, Internship at Shoes Factory in Misurata, Libya

https://www.researchgate.net/publication/344772515_Technical_report_of_maintenance_and_operation_internship_at_shoes_factory_in_Misurata_Libya

Mohamed Abuella, 2000 at Higher Center of Poly-Profession, Misurata, Libya

Electrical Operation & Maintenance for fulfilling requirement of the Higher Diploma



Research

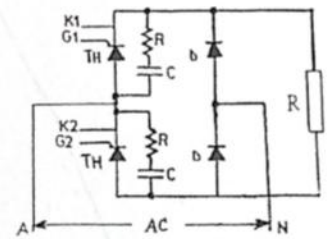
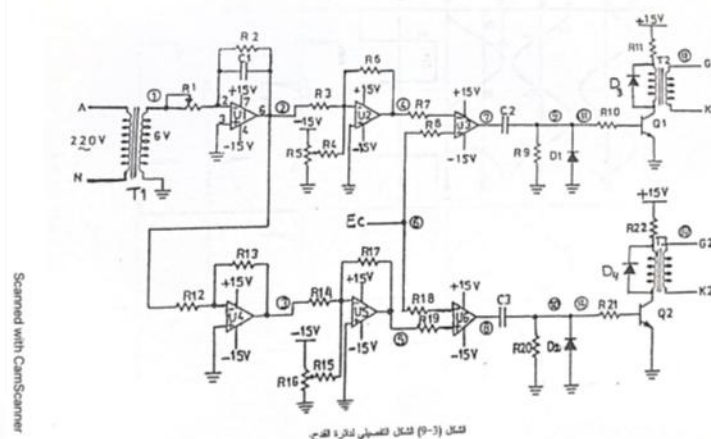
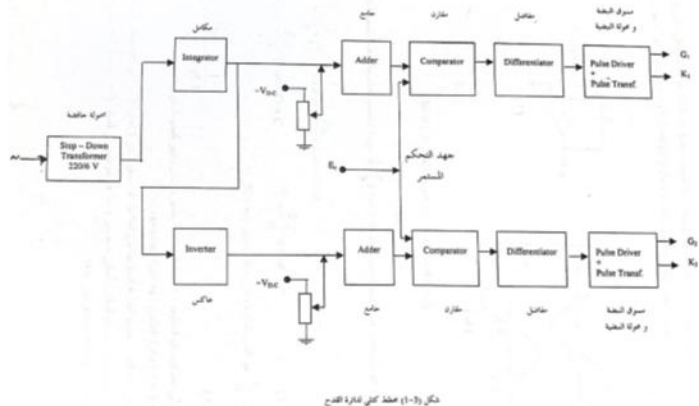
Triggering Circuit for SCR Thyristors of an AC-DC Converter

https://www.researchgate.net/publication/277109663_Triggering_Circuit_for_SCR_Thyristors_of_an_AC-DC_Converter

Mohamed Abuella, Ali Mohamed, Al Sayed Hamady, Advisor: Safa Samarmad
Tech Diploma Project, 2001 at Higher Center of Poly-Profession, Misurata, Libya

Higher Diploma project was in Power Electronics area. Since the task of the project of three-members-group was to build a triggering electronic circuit for a rectification bridge of Thyristors

Acquired Expertise: Electrical Wiring & Installations, Maintenance & Operation



Number: 30000
Status: Production
Package: TO-48 (14)
Type: Discrete
Case: DISCRETE
Type: SCR
Support: Datasheet
Doc: SCR Phase Control (discrete)
FOO # 30000



Parameter	Limit	Units	Condition	Value
VDRM	MIN	Volts	NA	1200
IT(RMS)	MAX	Amps	NA	35
IT(av) comp (a)	MAX	Amps	NA	22
@ TC	---	o C	NA	85
ITSM (50Hz)	MAX	Amps	NA	335
ITSM (60Hz)	MAX	Amps	NA	355
Vgt	Max	Volts	NA	2
Igt	Max	mAmps	NA	60
VTM comp (a)	MAX	Volts	NA	1.7
@ ITM comp (a)	---	Amps	NA	70
DVGT	MAX	V/us	NA	300
RRn(JC)	MAX	o C/W	NA	85

Research

Study of NEPLAN Software for Power Flow and Short Faults Analysis

https://www.researchgate.net/publication/277110587_Study_of_NEPLAN_Software_for_Load_Flow_and_Short_Faults_Analysis/stats

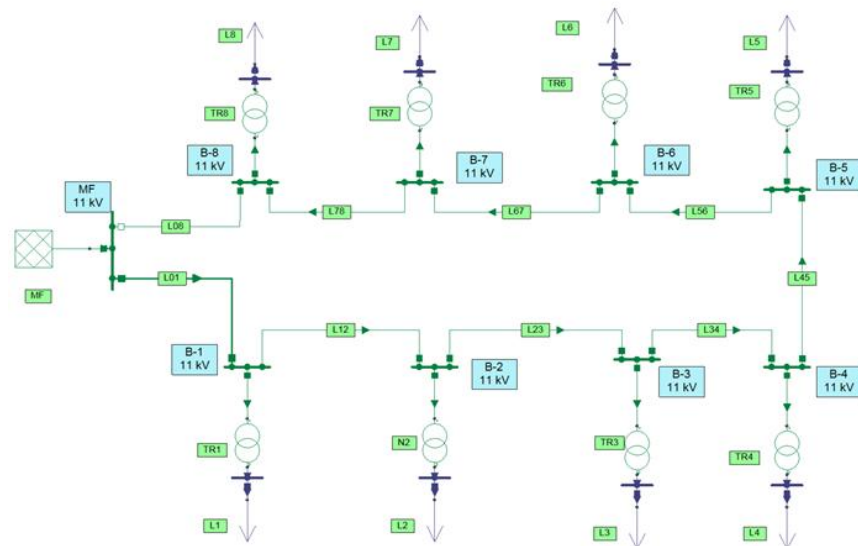


NEPLAN
Smarter Tools

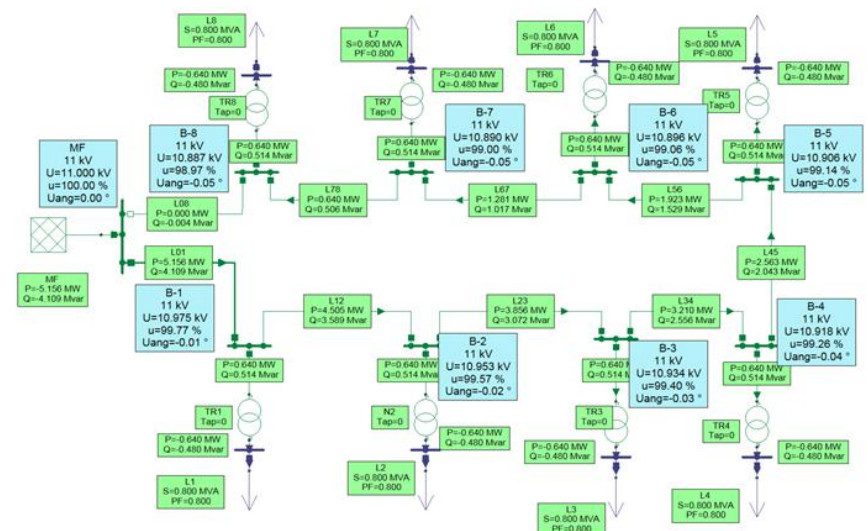
B.Tech Project, 2008 at College of Industrial Technology, Misurata, Libya
Advisor: Mohamed Shetwan

Acquired Expertise: Teaching, Tutorials, Lab Modeling & Simulations, Curriculum Revision & Preparation, Dedication, Listening, "Try to Modeling the Student's Way of Thinking."
Software Tools including: MS Office, MATLAB, NEPLAN, PLC's Ladder Logic

شبكة التوزيع الكهربائية (11/0.4 KV) للوحدات السكنية



• حساب سريان القدرة لشبكة التوزيع:



النتائج ظاهرة على مخطط شبكة التوزيع الكهربائية (11/0.4 KV) للوحدات السكنية

Research



SMART GRID, Seminar

<https://www.slideshare.net/MohamedAbuella/smart-grid-37661484>

Smart Grid Presentation in Seminar Course, 2012 at Southern Illinois University at Carbondale

Study of particle swarm for optimal power flow in IEEE benchmark systems including wind power generators

<https://www.proquest.com/openview/21da3b4335a4c23278e9bd91d67a7784/1?pq-origsite=gscholar&cbl=18750>

Master of Science Thesis, 2012 at Southern Illinois University at Carbondale, USA

Advisor: Constantine Hatziadoniu

Acquired Expertise: Power Systems Analysis, Operation and Planning, Systems Optimization, Smart Grid, Research Conducting, Software Tools: MATPOWER, PowerWorld, PSAT, LaTeX

Research

Study of particle swarm for optimal power flow in IEEE benchmark systems including wind power generators

<https://www.proquest.com/openview/21da3b4335a4c23278e9bd91d67a7784/1?pq-origsite=gscholar&cbl=18750>



Master of Science Thesis, 2012 at Southern Illinois University at Carbondale, USA

Advisor: Constantine Hatziaodoniu

$$J_{Min} = \sum^M C_i(p_i) + \sum^N C_{wi}(w_i) + \sum^N C_{p,i}(w_i) + \sum^N C_{r,i}(w_i)$$

Subject to: **Where:** $C_i = a_i P_i^2 + b_i P_i + c_i$

$$p_{i,min} \leq p_i \leq p_{i,max}$$

$$0 \leq w_i \leq w_{r,i}$$

$$\sum_i^M p_i + \sum_i^N w_i = L$$

$$V_i^{min} \leq V_i \leq V_i^{max}$$

$$S_{line,i} \leq S_{line,i}^{max}$$

$$C_{w,i} = d_i w_i$$

$$C_{p,i} = k_{p,i} \int_{w_i}^{w_{r,i}} (w - w_i) f_W(w) dw \text{ (underestimation)}$$

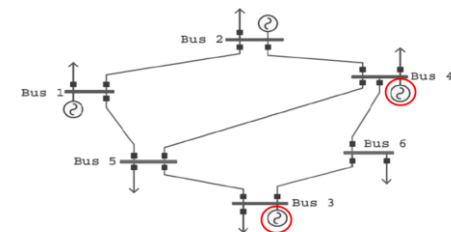
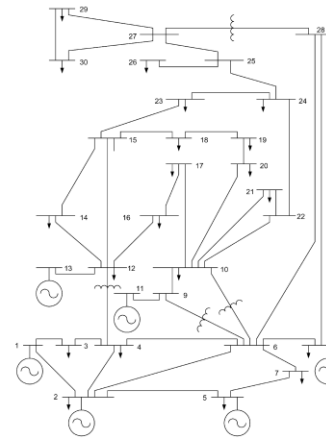
$$C_{r,i} = k_{r,i} \int_0^{w_i} (w_i - w) f_W(w) dw \text{ (overestimation)}$$



$$C_i = a_i P_i^2 + b_i P_i + c_i$$



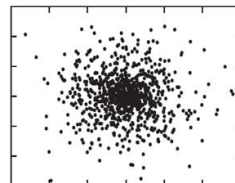
$$C_{w,i} = d_i w_i$$



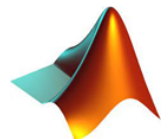
Gen. No.	a (\$/MW ² .hr)	b (\$/MW.hr)	c	P _G . low (MW)	P _G . high (MW)
1	0.012	12	105	50	250
2	0.0096	9.6	96	50	250
3	0	8	0	0	40
4	0	6	0	0	40



PSO



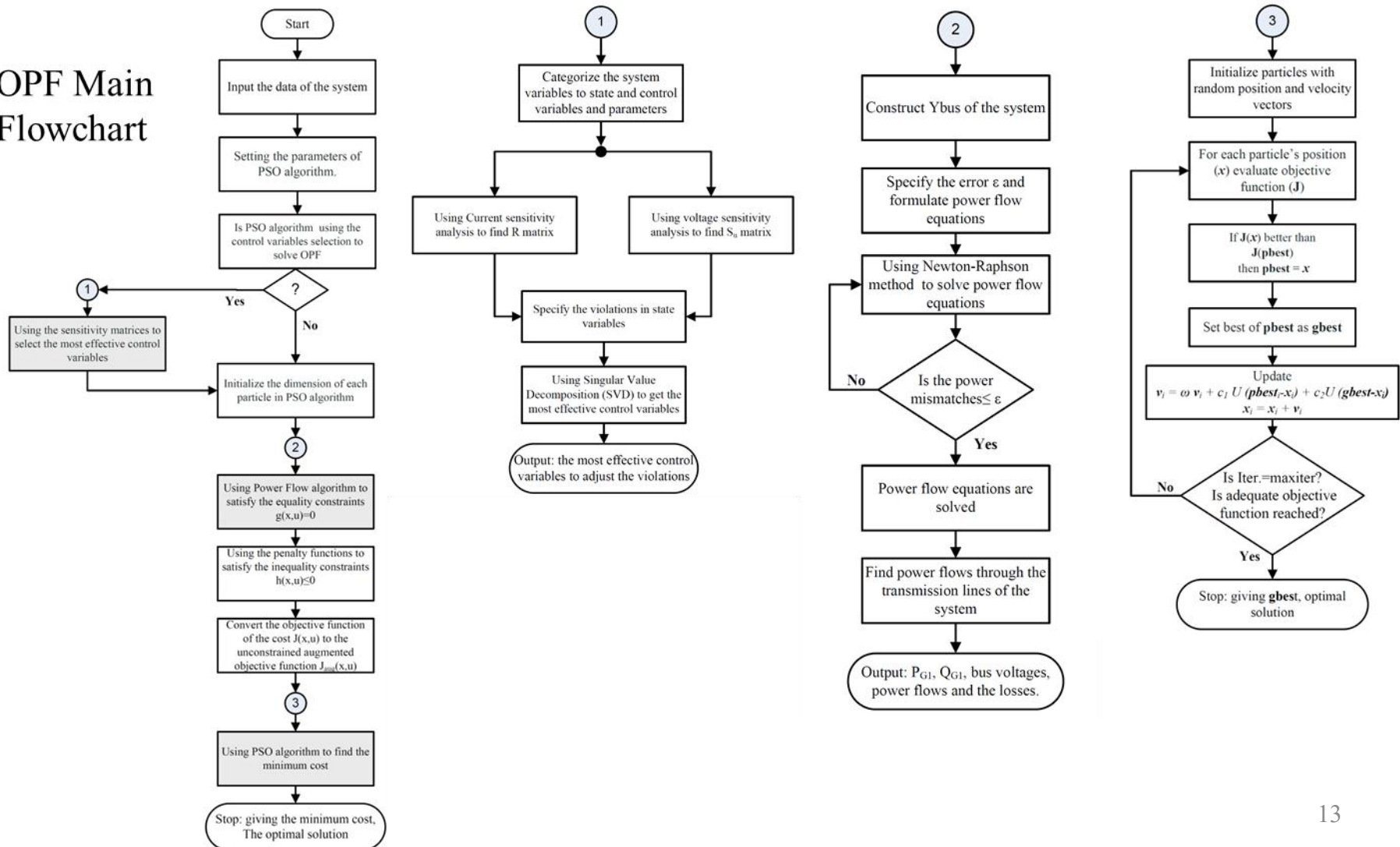
Particle Swarm Optimization (PSO) algorithm is used for solving this optimization problem.



Research

Study of particle swarm for optimal power flow in IEEE benchmark systems including wind power generators

OPF Main Flowchart



Research

A Post-Processing Approach for Solar Power Combined Forecasts of Ramp Events

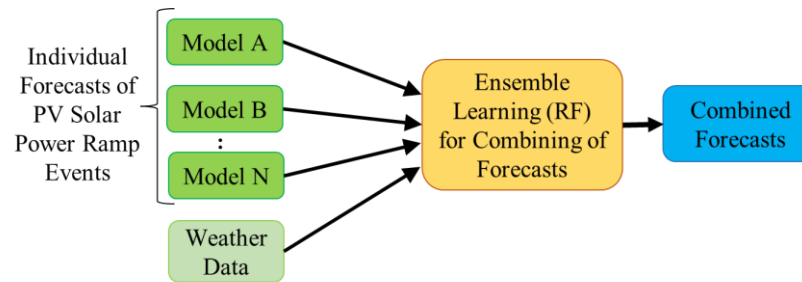
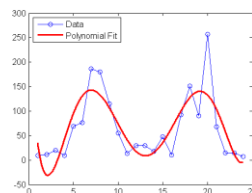
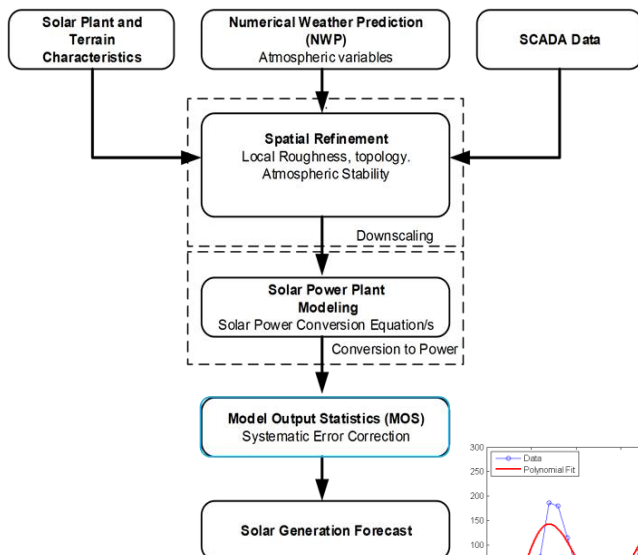
<https://www.proquest.com/openview/42049145119c7760f93ea736b37a0930/1.pdf?pq-origsite=gscholar&cbl=18750>

PhD Thesis, 2018 at University of North Carolina at Charlotte, USA

Advisor: Badrul Chowdhury



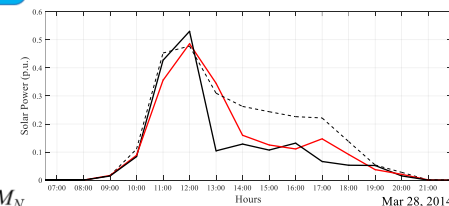
Acquired Expertise: Energy Analytics, Energy Markets, Renewable Energy Integration, Asset & Supply Chain, Time Series Analysis & Modeling, Risk & Uncertainty Quantification, Machine Learning, Big-Data Processing, Research Publishing & Peer Reviewing, Software Tools including SAS, R, and Python



General diagram of combining different models

$$F_{comb} = W_A * M_A + W_B * M_B + W_C * M_C + \dots + W_N * M_N$$

Method of Combining The Models $\Rightarrow$ Random forest (RF) is chosen to be the **ensemble learning** method for combining the various models' outcomes.



Why Forecast?

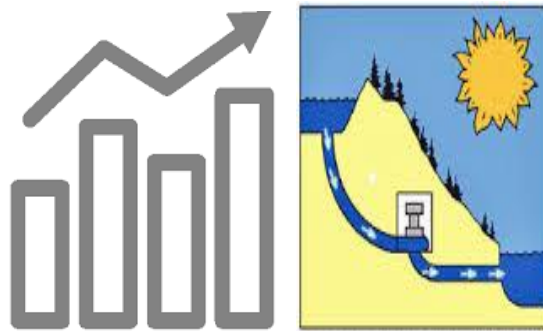
Research

$$P_{\text{Supply}} = P_{\text{Demand}} + P_{\text{Loss}}$$

**PV Solar Power
Generations
are Too Variable**



**Coordination with Operating
Reserves and Energy Storage
Systems**



**Reducing
Cost
and Pollution**

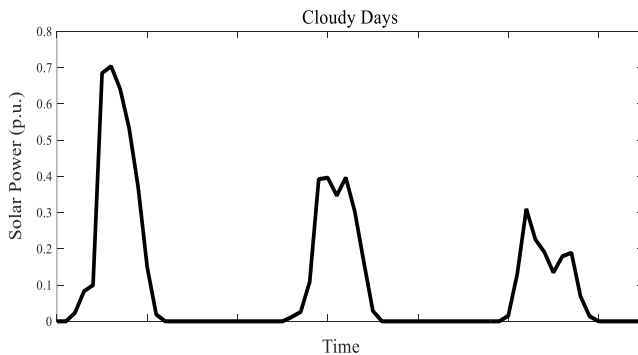


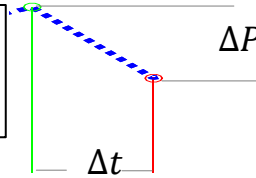
Illustration of the motivation of PV solar power forecasts

Research

Definition of Ramp Events

Solar power ramp rate (RR) is *the change of solar power during a certain time interval*.

$$\text{Ramp Rate, } RR(t) = \frac{dP(t)}{dt} = \frac{P(t + D) - P(t)}{D}$$



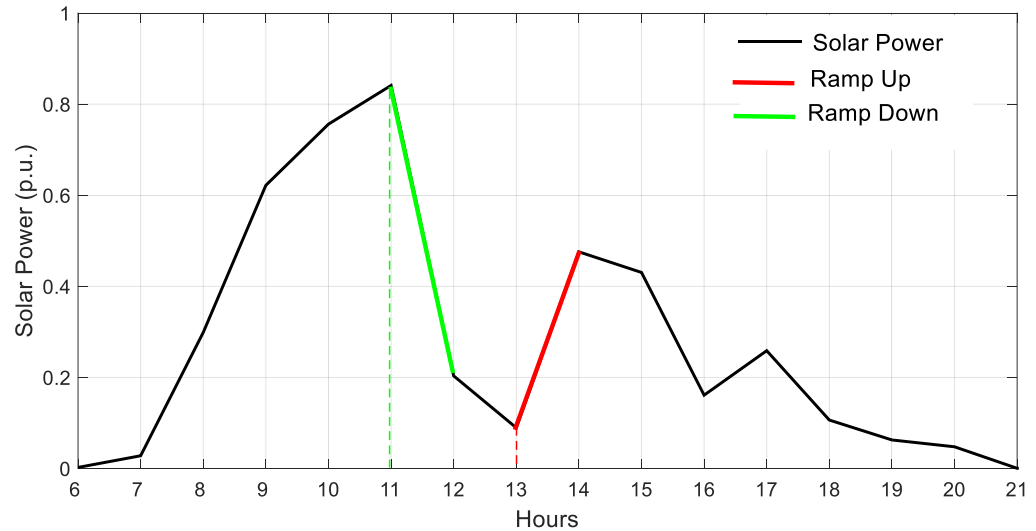
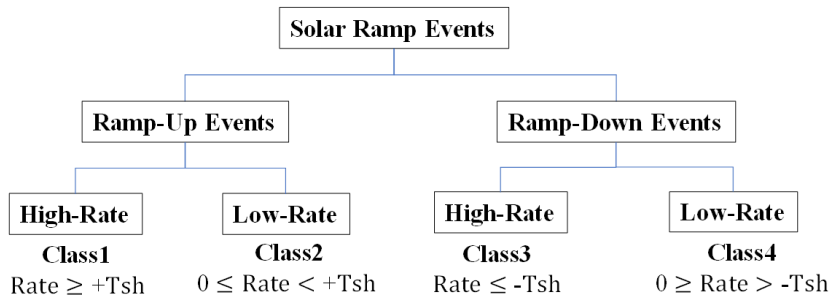
where $P(t)$ is the solar power of the target hour, it can also be its forecast $F(t)$; D is the time duration for which the ramp rate is determined.

For the illustrated cloudy day below:

Ramp rate, $\frac{\Delta P}{\Delta t} = \frac{0.2 - 0.85}{12:00 - 11:00} = -0.65$ (-65%) ramp down of its normal capacity, (pu/hr)

Ramp rate, $\frac{\Delta P}{\Delta t} = \frac{0.48 - 0.1}{14:00 - 13:00} = +0.38$ ($+38\%$) ramp up of its normal capacity, (pu/hr)

Some ramps are with low rates, while others with high rates.



Distribution of the classes of solar power ramp events

Ramp Events During a Cloudy Day

Research

Potential Applications

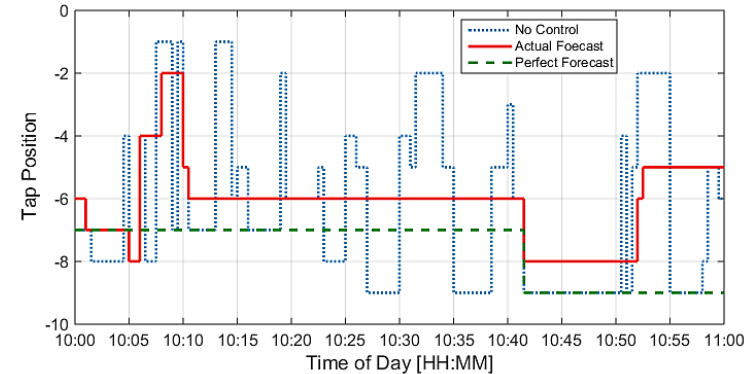
There are several applications of power systems that rely on solar power ramp event forecasts

Distribution level:

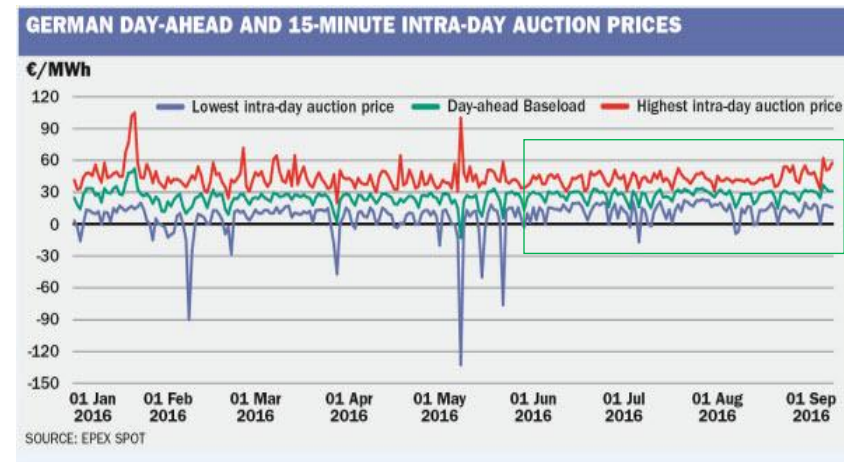
- Optimizing the voltage regulation equipment.
- Control schemes of energy storage systems.

Transmission / bulk level:

- Trading & dispatching the operating reserve.
- Managing the ramp capability / system flexibility with high-level of renewable energy integration.

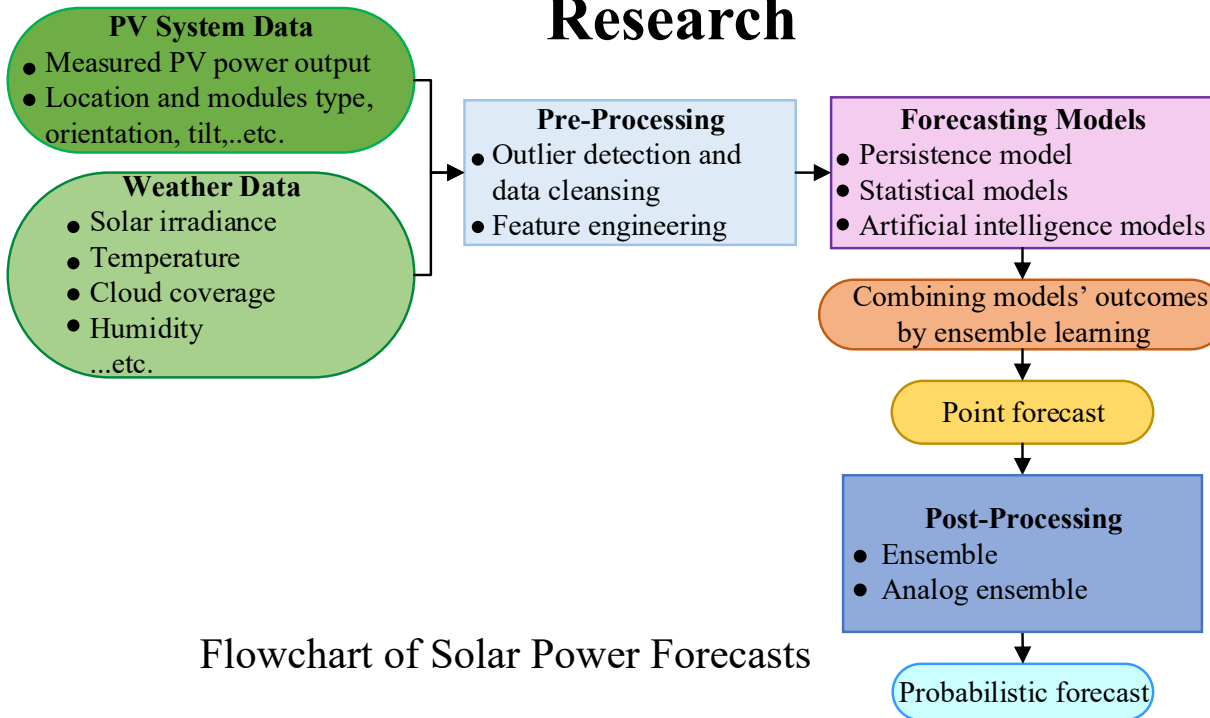


Optimizing the Transformer's Tap Changer position sequences using the solar forecast

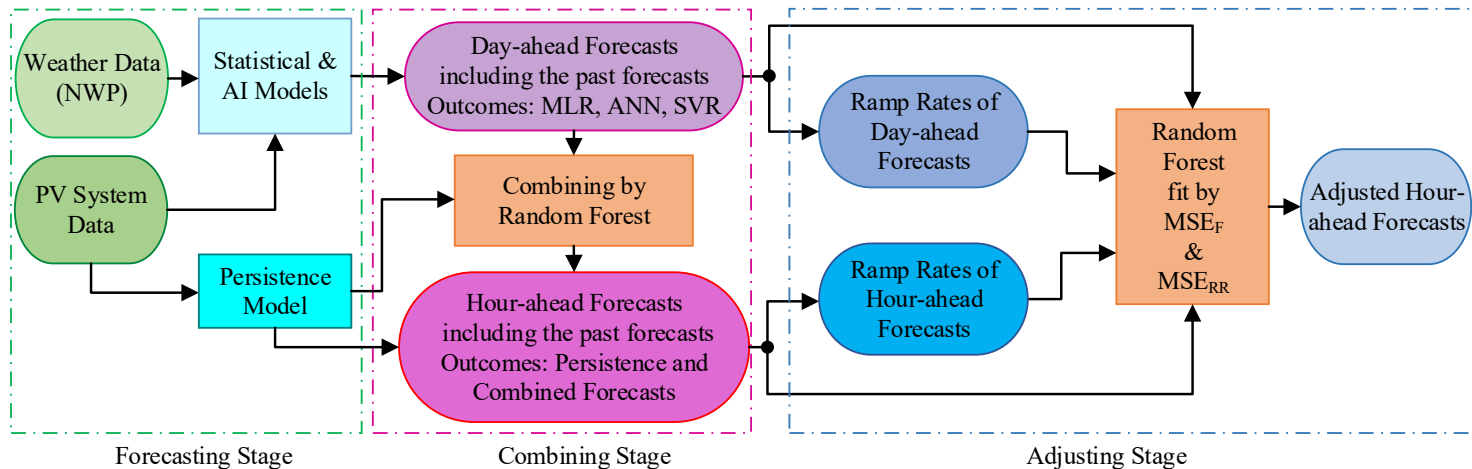


EPEX: European power exchange spot trading

Research

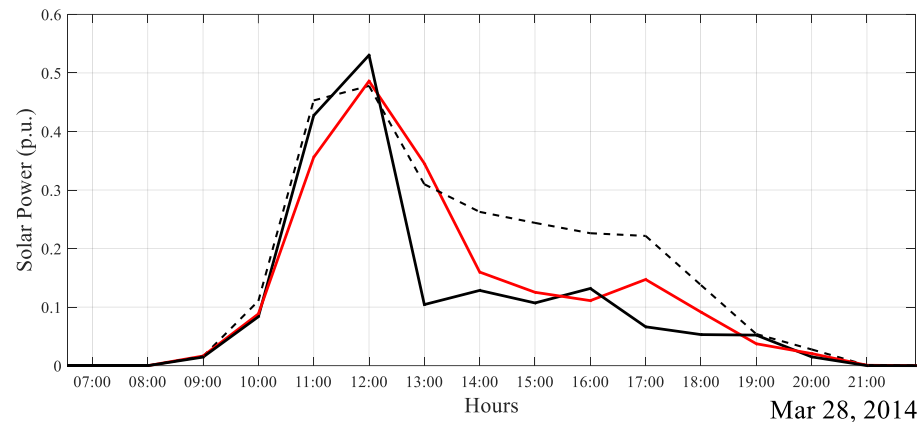
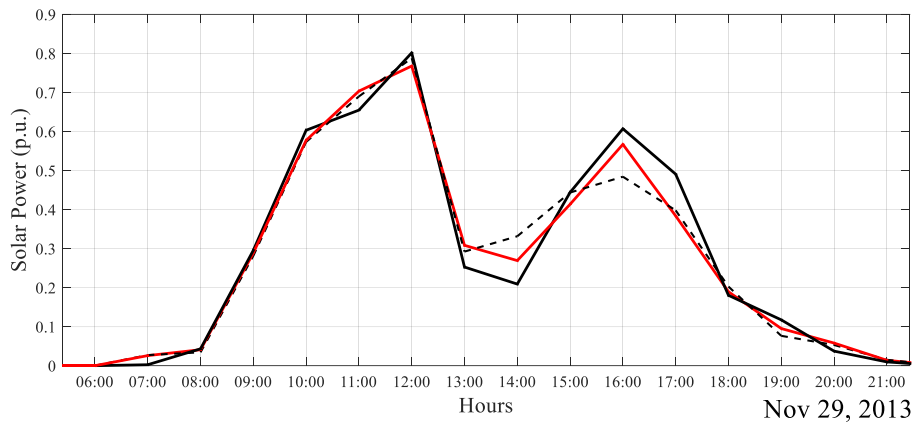
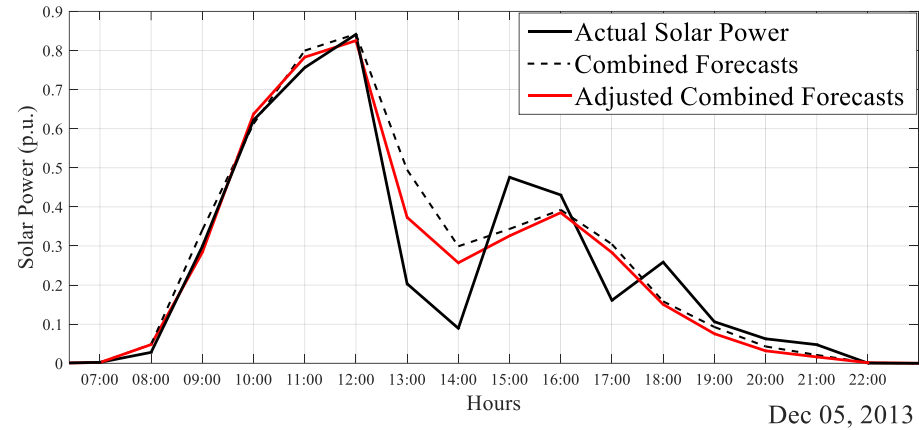
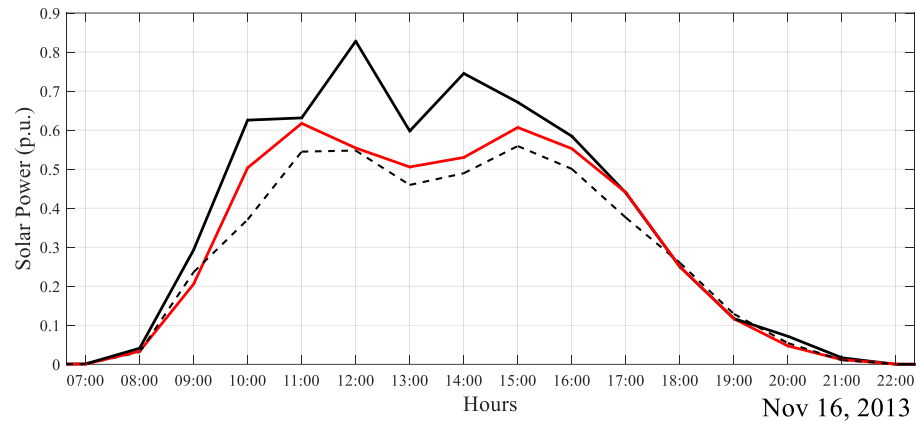


Flowchart of Solar Power Forecasts



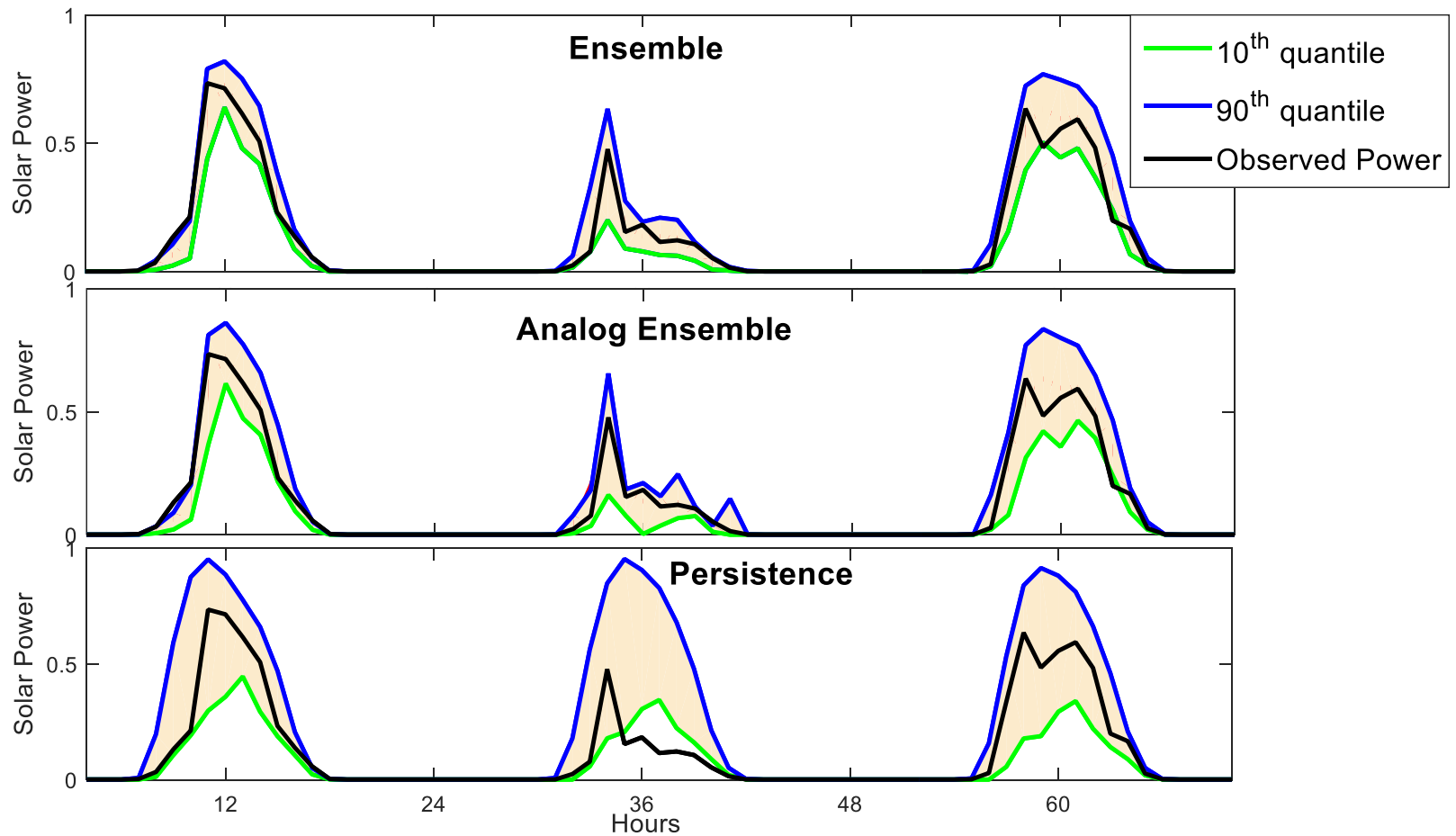
Block diagram of the adjusting approach

Research



Combined forecasts of solar power for cloudy days before and after applying the adjusting

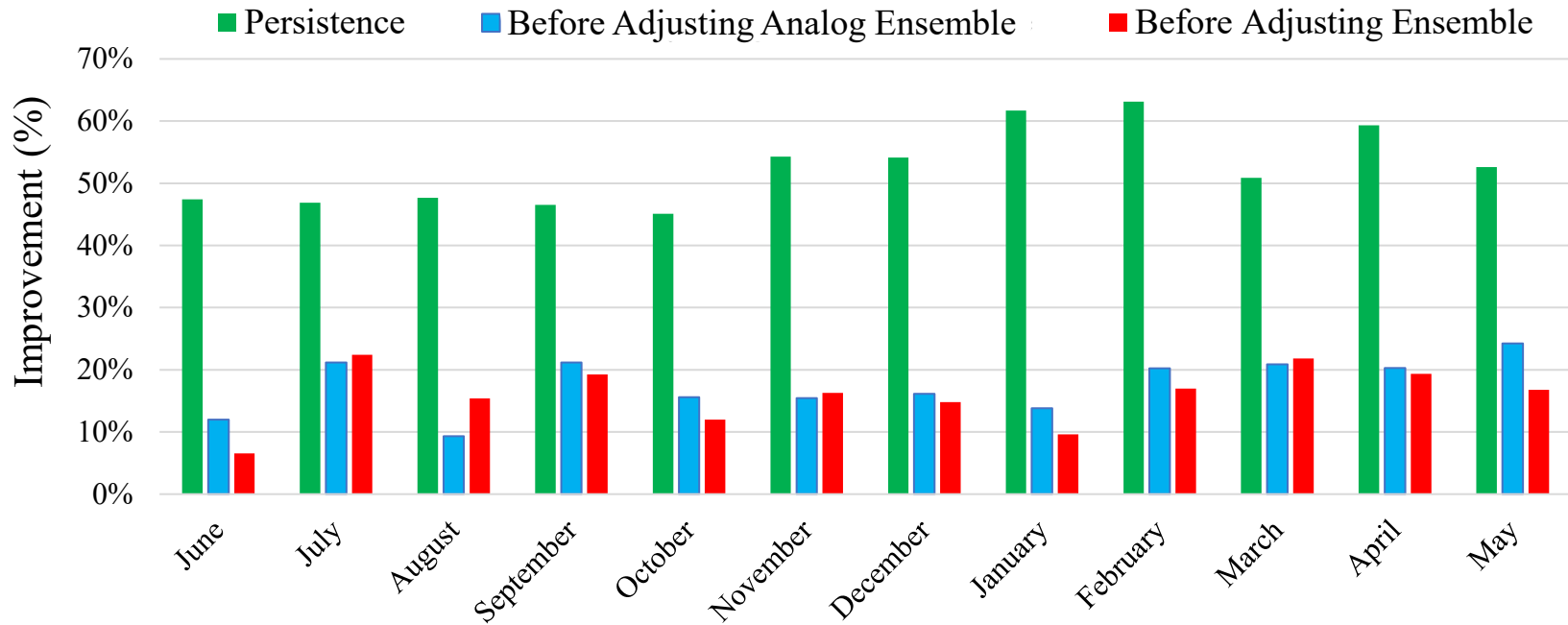
Research



Graphs of the probabilistic forecasts of the three methods for three days

Research

Improvement of Adjusted Ensemble-based Probabilistic Forecasts Over:



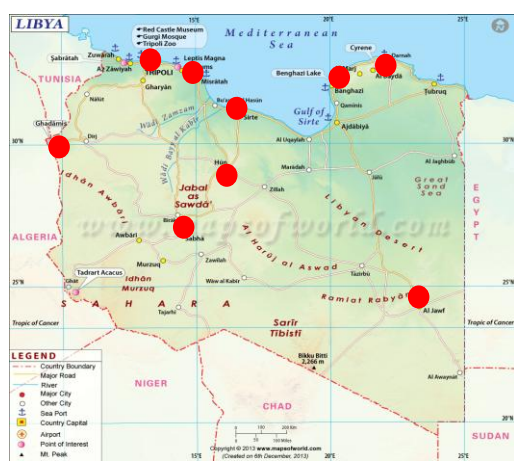
$$\text{Skill Score (\%)} = \left(1 - \frac{\text{Metric}_{\text{method}}}{\text{Metric}_{\text{reference}}} \right) * 100$$

Average Improvement of the Adjusting Approach:
51% over Persistence;
16% over Analog Ensemble;
15% over Ensemble.

Research

Planning and Analysis for Solar Energy in Libya

9 Locations for Comparison of Solar Energy Modeling and Analysis: Tripoli, Misurata, Sirte, Benghazi, Derna, Houn, Gadamis, Sabha, Kufra
 Typical Meteorological Year (TMY) data represents the weather for a "median year". <https://developer.nrel.gov/>

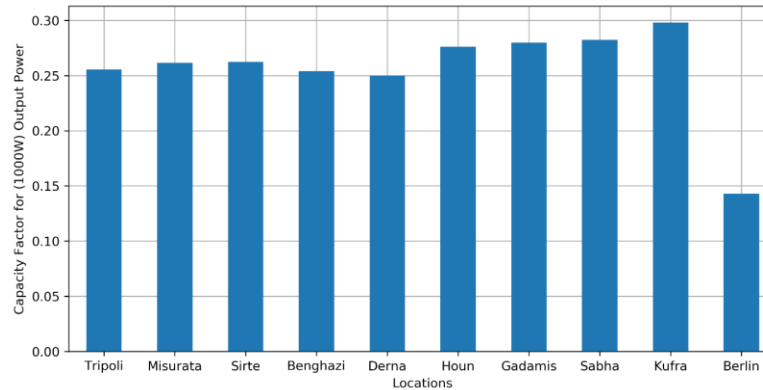


Solar Energy Modeling

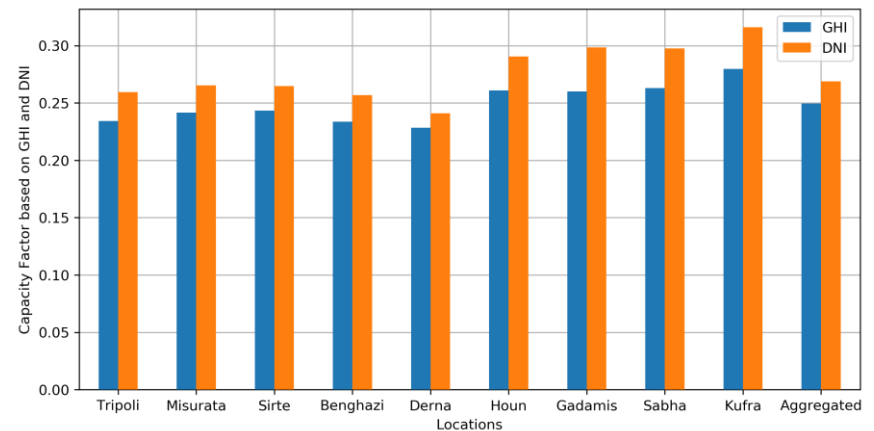
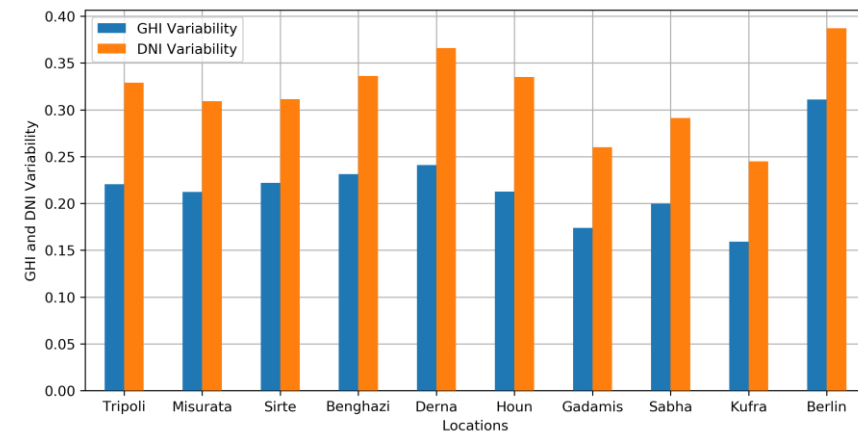
Capacity Factor of Solar Energy Resources

Variability of Solar Energy Resources

Aggregation of Solar Energy Resources



The Capacity Factor is calculated based on output power (PWac).
 The Rating of Solar PV System = 1000W during an entire year = 8760 hours.



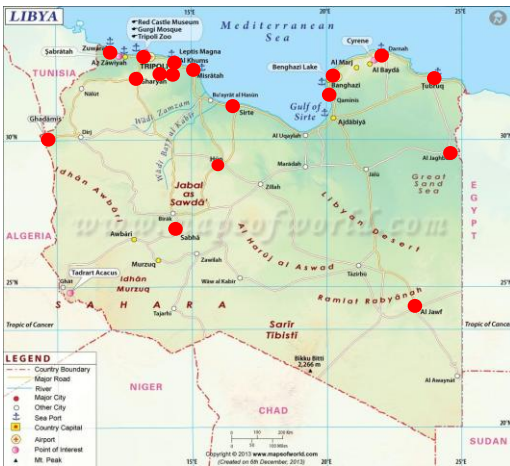
Research

Planning and Analysis for Wind Energy in Libya

17 Locs: Tripoli, Misurata, Tarhunah, Ghanima, Msallata, Zuwara, Gharyan, Sirte, Benghazi, Magrun, Derna, Houn, Gadamis, Sabha, Kufra, Tobruk, Jaghub

Typical Meteorological Year (TMY) data represents the weather for a "median year". Data are retrieved from NREL's Developer Network:

<https://developer.nrel.gov/> Comparison of Monthly Average Wind Speed (m/s) at Height of 10m

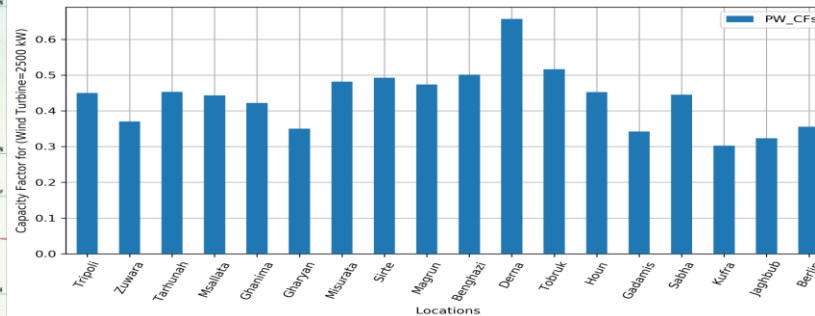


Wind Energy Modeling

Capacity Factor of Wind Energy Resources

Variability of Wind Energy Resources

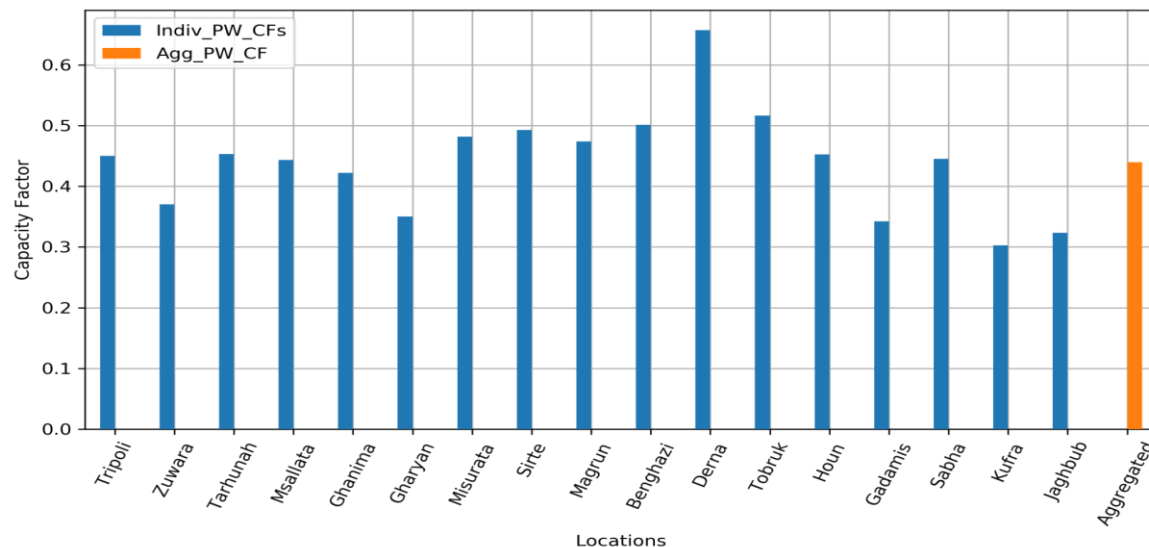
Aggregation of Wind Energy Resources



The Capacity Factor is calculated based on Output power (PWac).

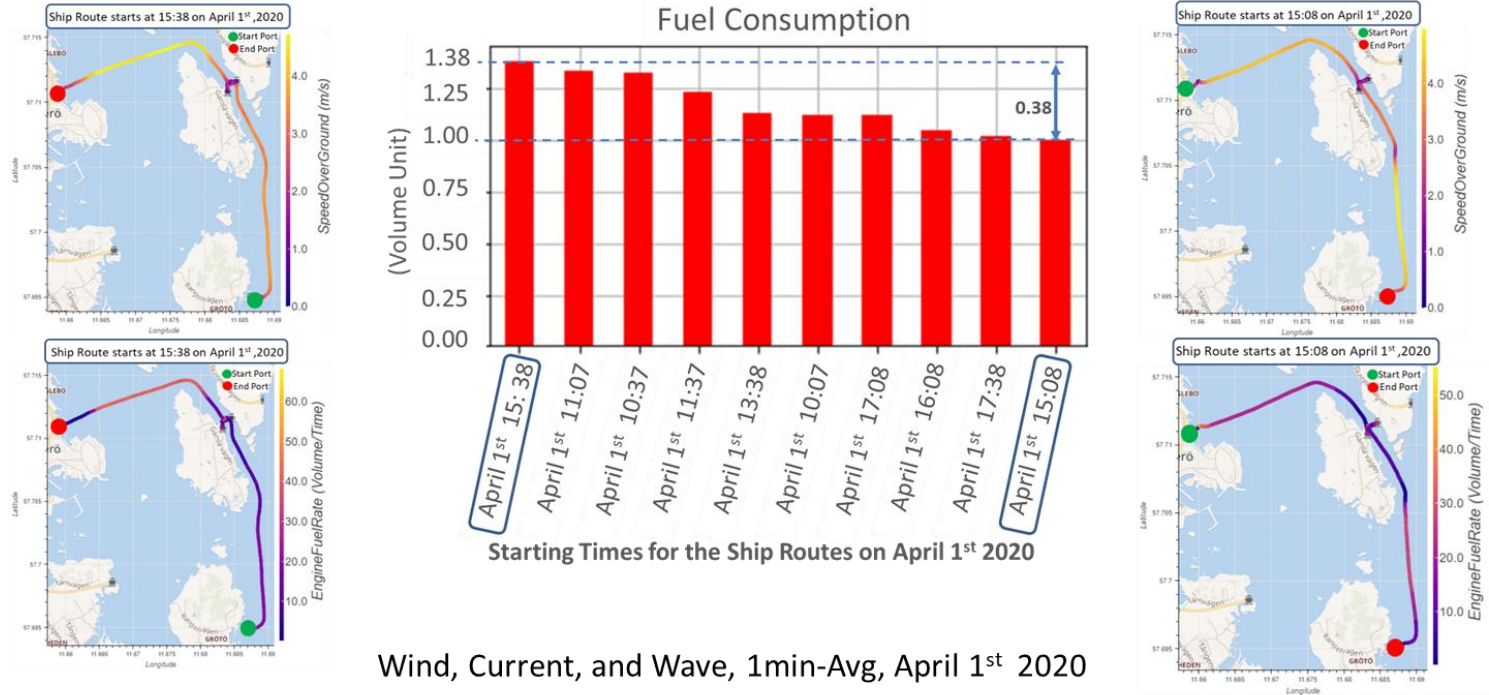
The Rating of Solar PV System = 1000W during an entire year = 8760 hours.

Berlin in Germany has been added just for sake of comparison.

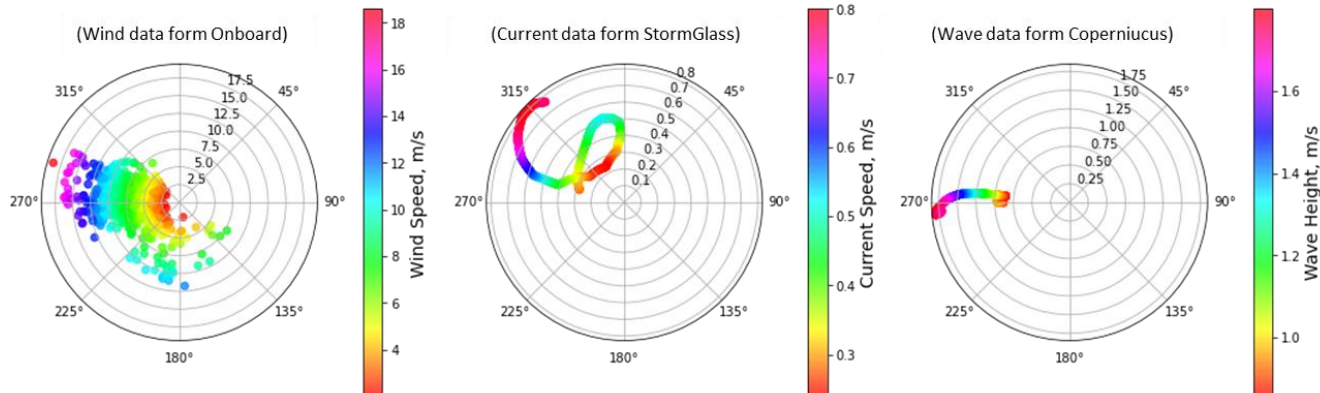


Research

Research Work for Improving the Vessel's Energy Efficiency



Wind, Current, and Wave, 1min-Avg, April 1st 2020



Research

Data Analytics for Improving the Vessel's Energy Efficiency

Descriptive Modeling

Input: Data of vessel's operational and environmental data, from onboard and external sources.

Outcome: Dataset for training and validation the predictive and prescriptive models.

Predictive Modeling

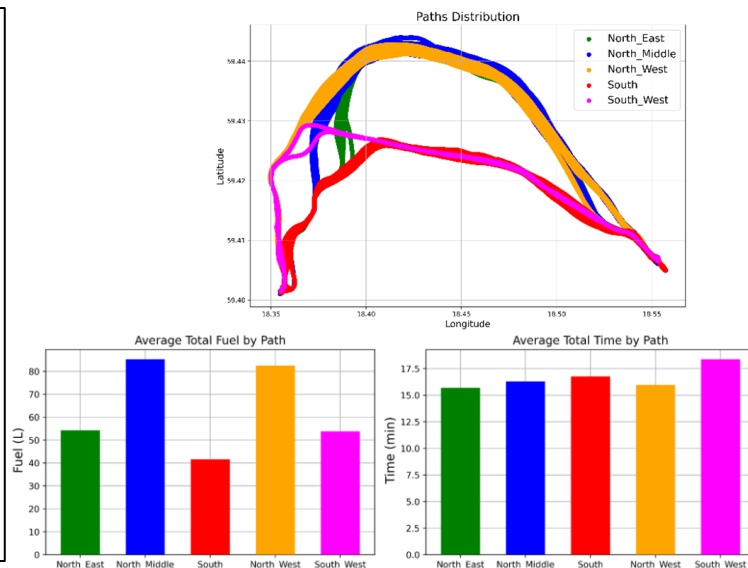
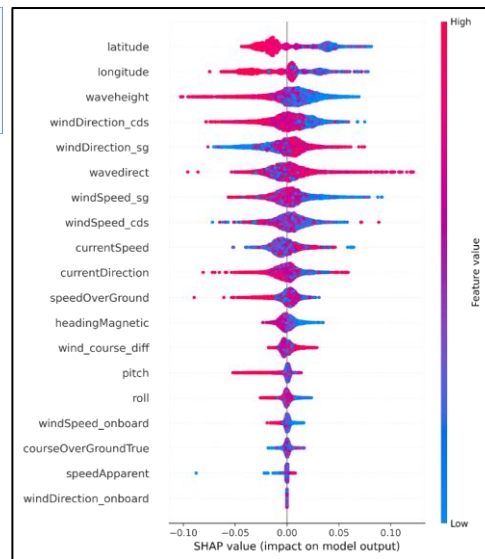
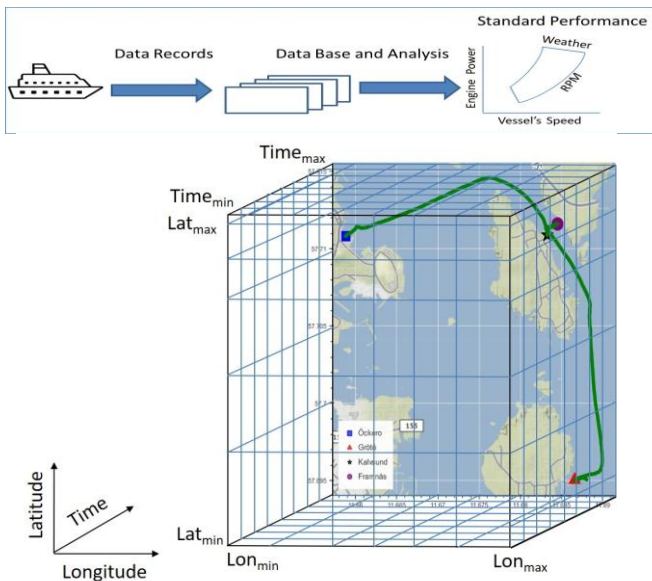
Input: Preprocessed data include operating and weather variables, such as vessel's speed and course, wind, wave, current, etc.

Outcome: Validated predictive models for fuel, distance and time.

Prescriptive Modeling

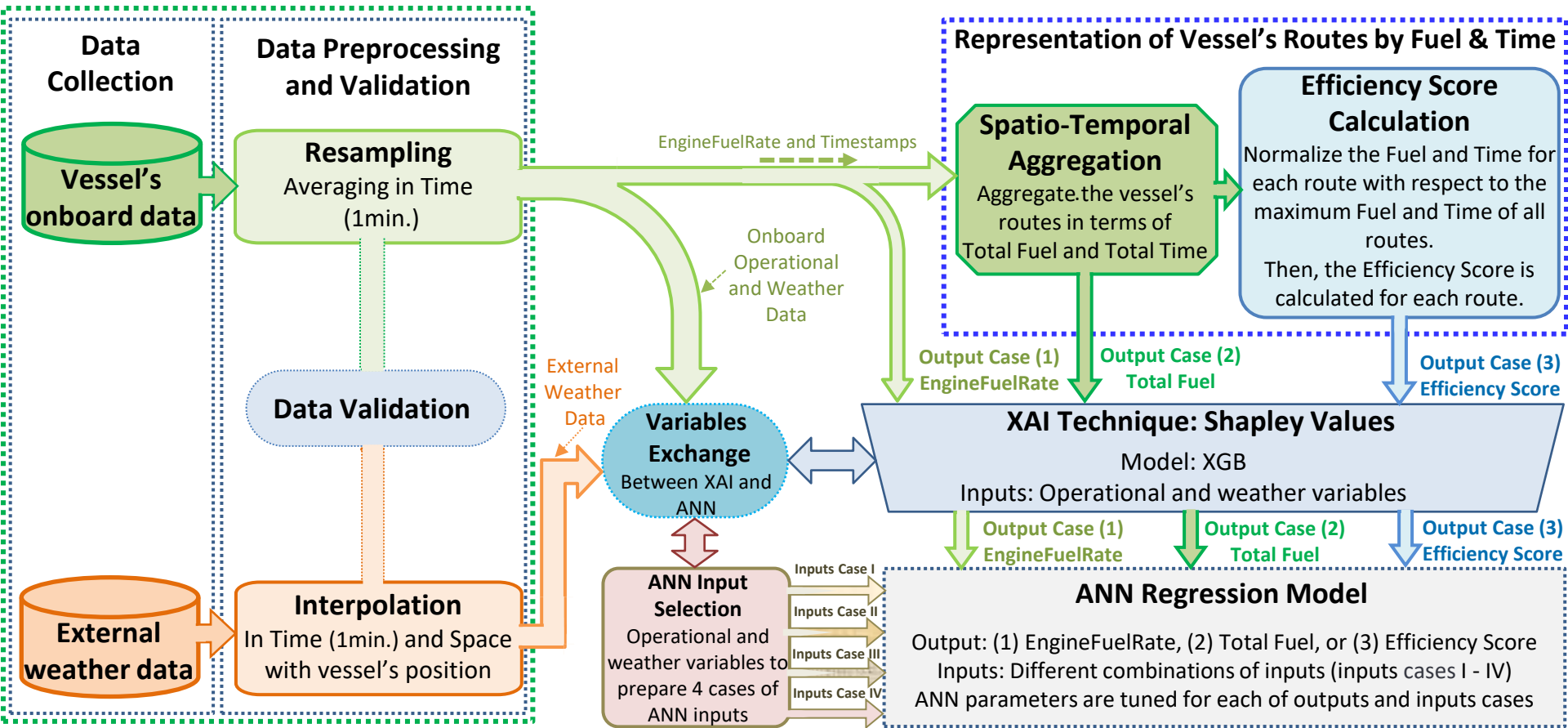
Input: Using solving algorithms, with control variables, such as vessel's speed and course, to find the optimal fuel and time.

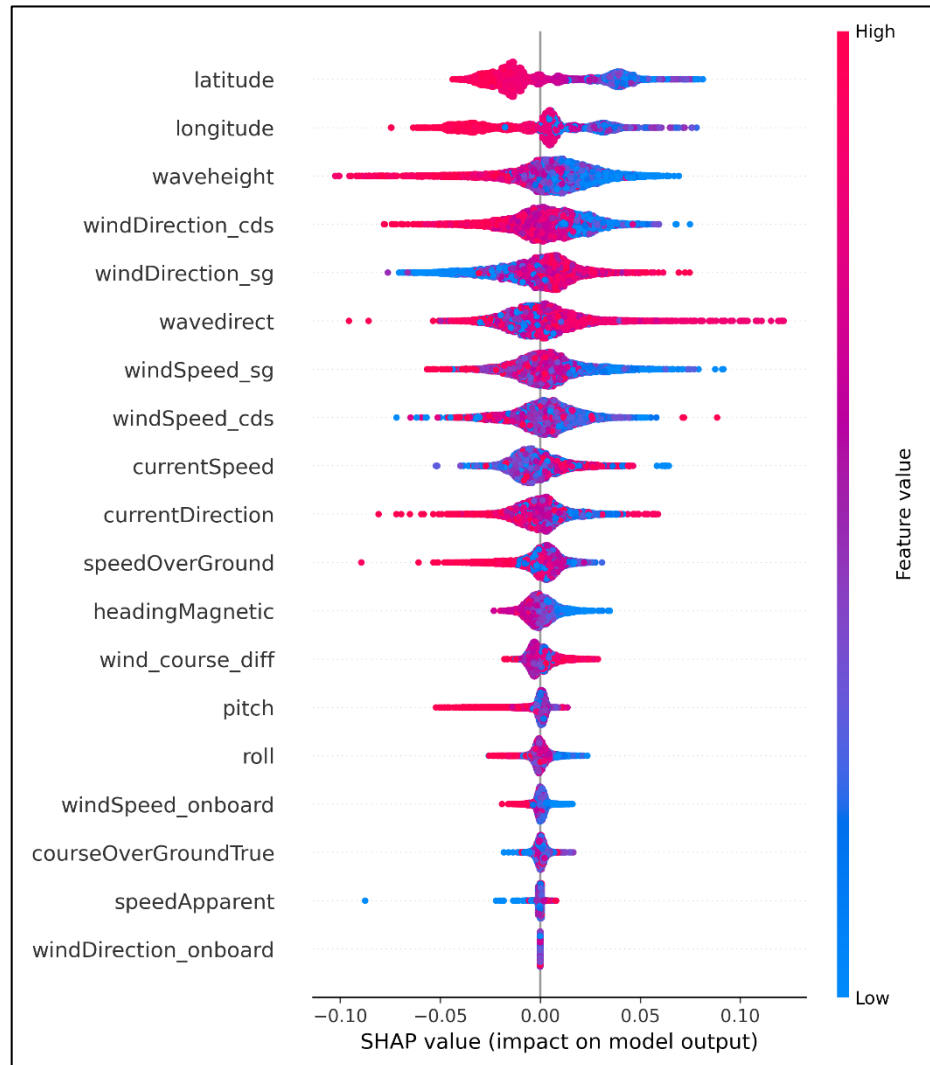
Outcome: Improving fuel consumption and meeting the operational conditions



Research

Workflow of Applying XAI for Improving the Vessel's Energy Efficiency





Shapely values for Regression of Eff-Score (Global, where Fuel and Time are normalized based on values of all routes)

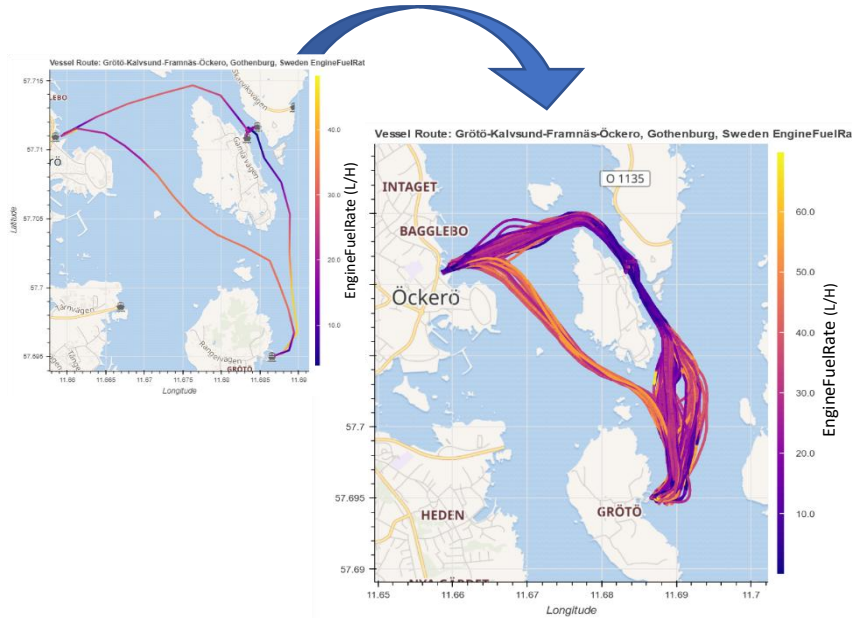
The spatio-temporal aggregation and using the efficiency score (Global) leads to the **causality**

Efficiency Score

It is calculated once for the entire route.

Research

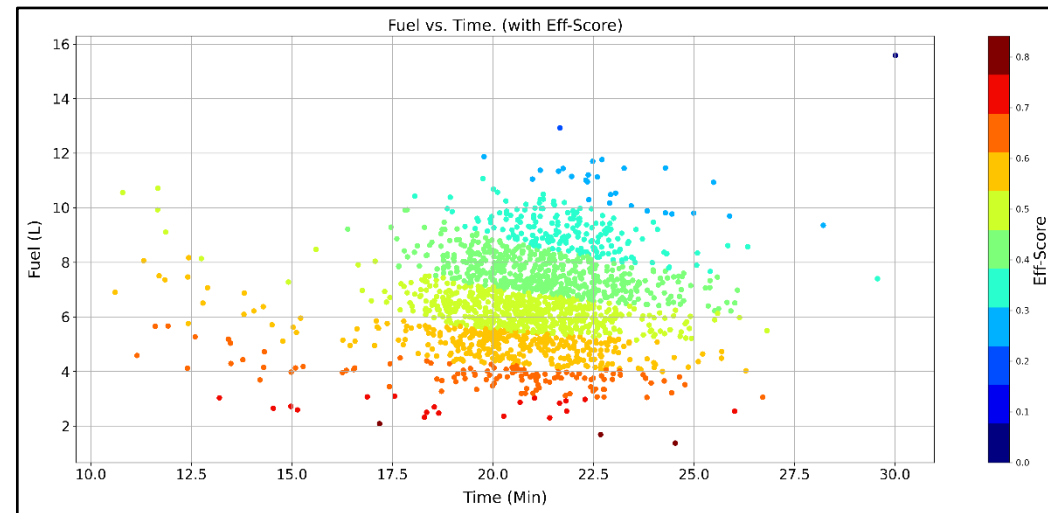
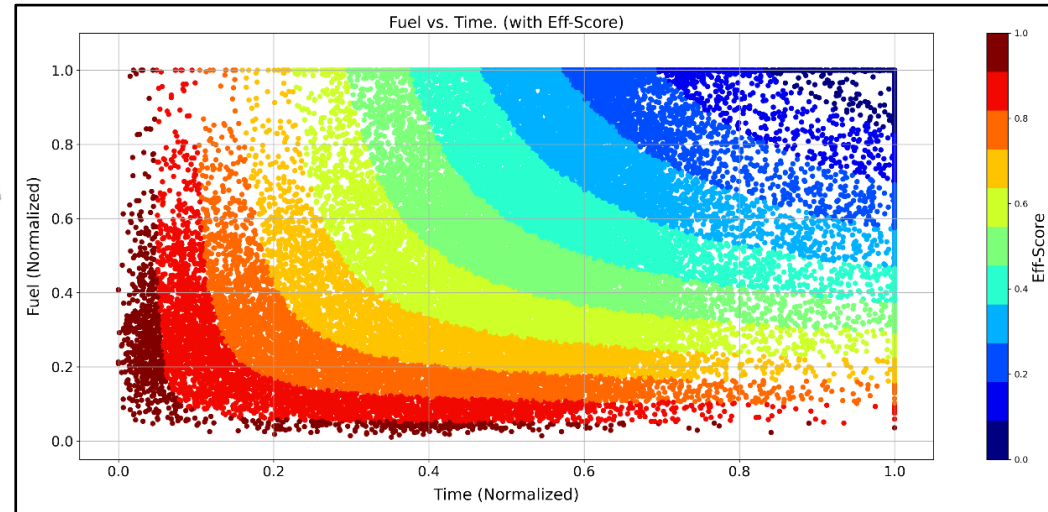
Predictive Analytics



1754 sequences (Routes)

$$Eff_{Cost} = \frac{2 * (Fuel\ Total_{norm} * Time\ Total_{norm})}{(Fuel\ Total_{norm} + Time\ Total_{norm})}$$

$$Eff_{score} = 1 - Eff_{Cost}$$



Research

Problem Formulation

Objective Function:
Minimizing the fuel consumption

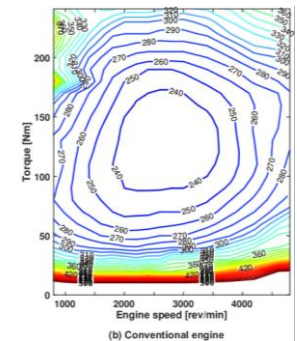
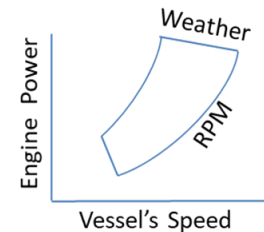
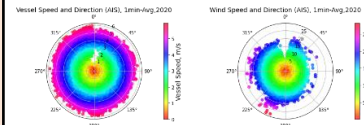
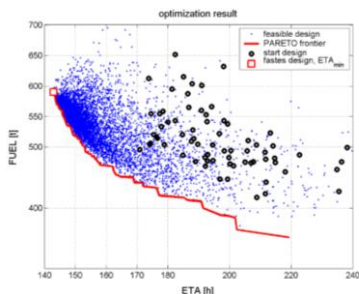
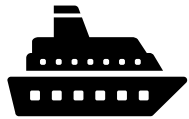
Solutions Finding

Solving algorithm:
Modeling and managing the engine power at any weather conditions by using a fuel estimation model

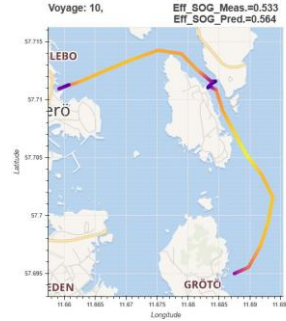
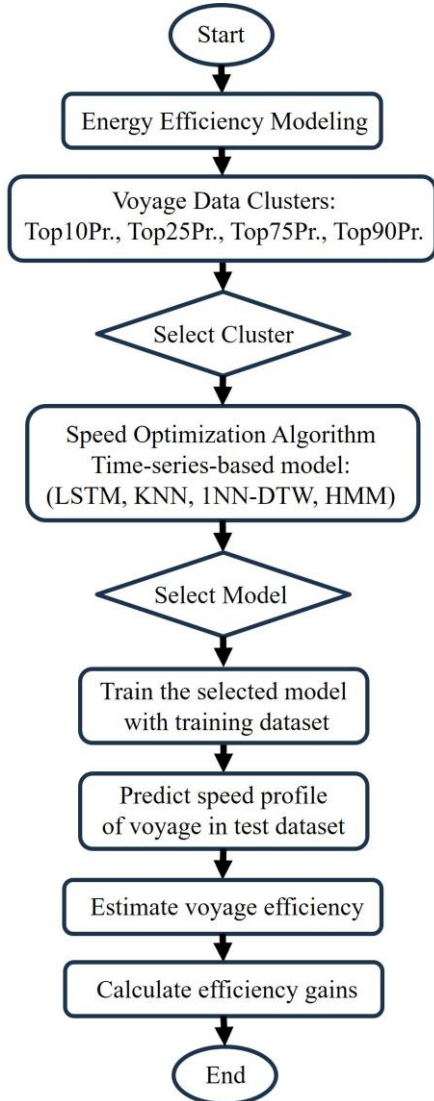
Control variables:
Ship's speed and course

Constraints:
Arrival time, geographic, safety, route smoothness, the ship's roll, and the engine power

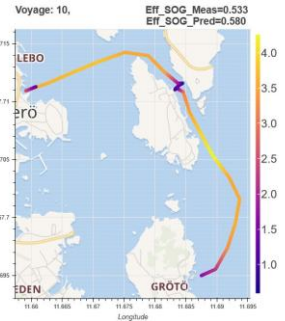
Objective:
Minimum fuel consumption



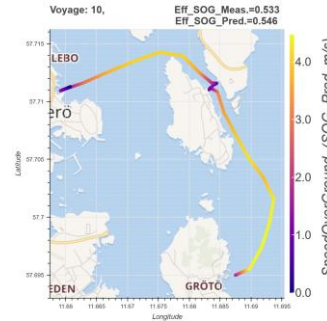
Framework of vessel voyage optimization



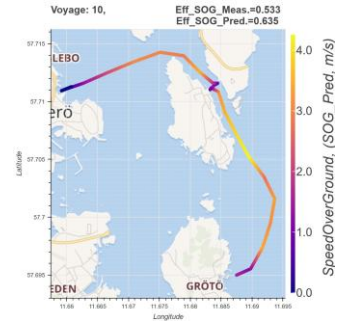
(a) LSTM-based model



(b) KNN-based model



(c) 1NN-DTW-based model



(d) HMM-based model

Cluster	Efficiency Score	LSTM	KNN	1NN-DTW	HMM
Top10Pr	Eff. Gains (%)	2.61	2.13	3.20	6.05
	IV Count (#)	134	114	127	139
Top25Pr	Eff. Gains (%)	2.38	1.58	3.23	1.30
	IV Count (#)	129	107	128	107
Top50Pr	Eff. Gains (%)	0.97	0.98	2.58	7.34
	IV Count (#)	100	106	117	140
Top75Pr	Eff. Gains (%)	-0.84	0.50	2.28	9.31
	IV Count (#)	60	93	119	141
Average	Eff. Gains (%)	1.28	1.30	2.82	6.00
	IV Count (#)	105.75	105.00	122.75	131.75

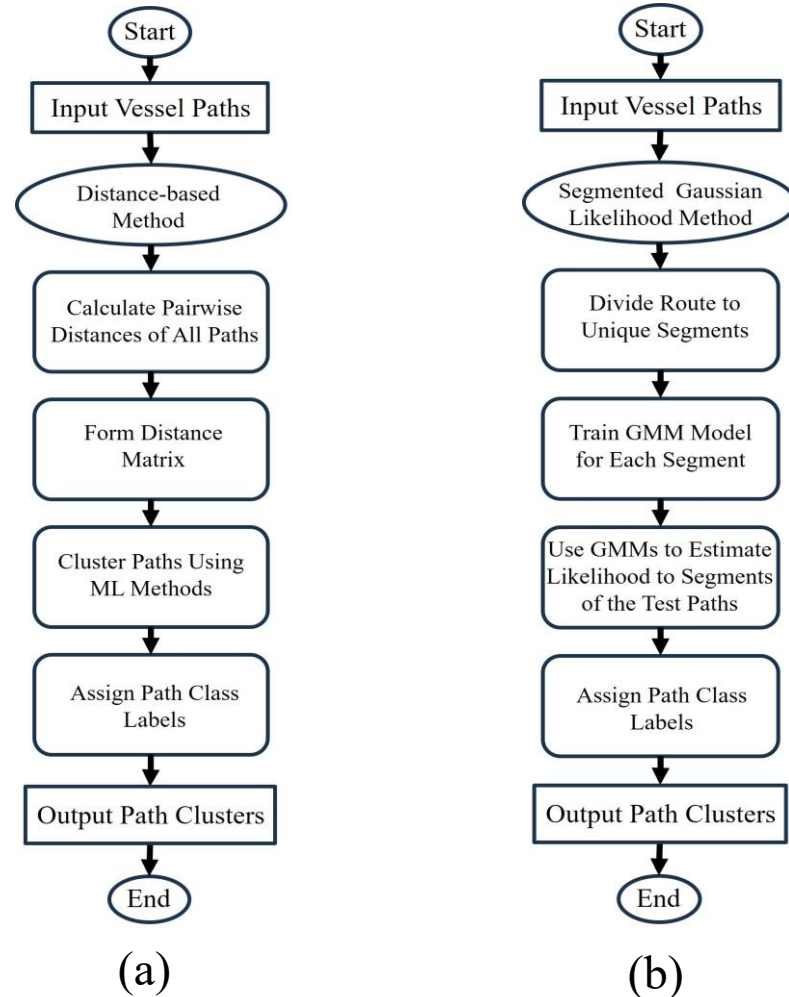
where,

$$Eff.Gain = \frac{Eff.Score_{Pred.} - Eff.Score_{Meas.}}{Eff.Score_{Meas.}} \times 100$$

Counts of improved voyages (IV Count#) out of 162 voyages in the test dataset

Research

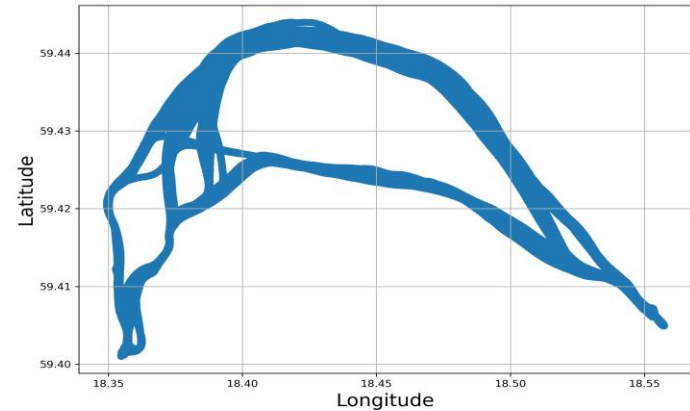
Spatial Clustering Approach for Vessel Path Identification



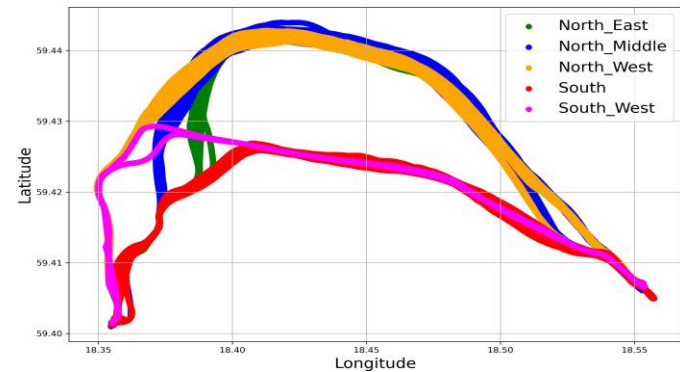
Framework of vessel path identification.

(a) Flowchart of distance-based method.

(b) Flowchart of segmented Gaussian likelihood method.



The vessel route

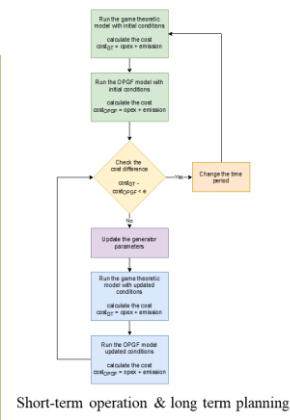
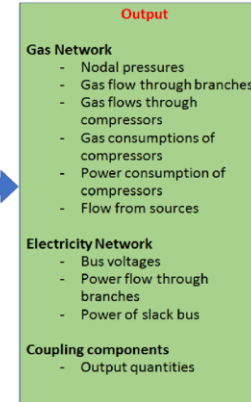
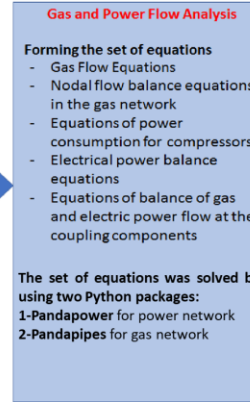
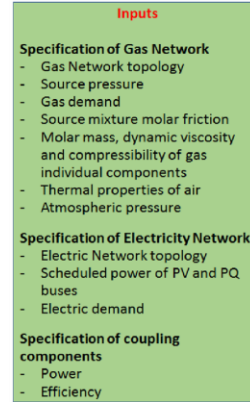
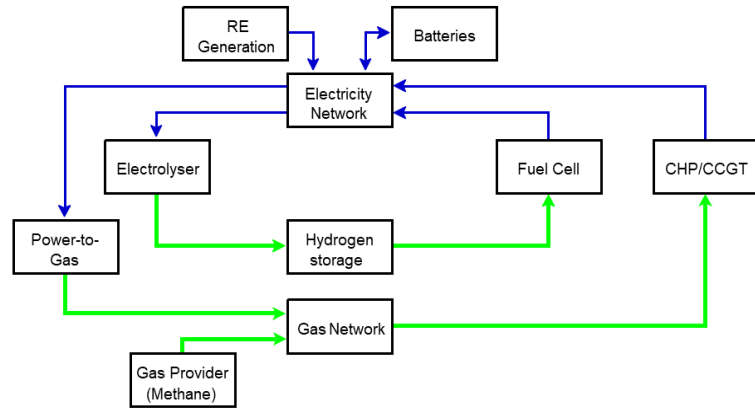


Main classes of path

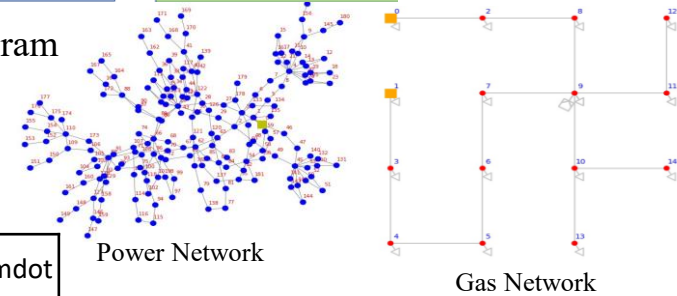
Research

Multi-Energy Model

HI ACT: Hydrogen Integration for Accelerated Energy Transitions



Schematic Diagram



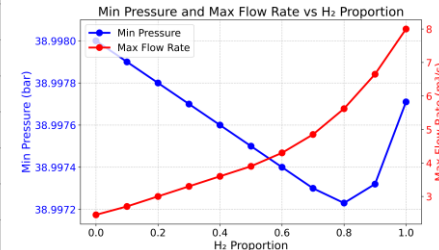
Framework for the Multi-Energy System

H2_prop	Total_cost	Optimal_q_e	Optimal_q_g	qg_H2	qg_gas	H2_mdott	Gas_mdott	Source_mdott
	(£)	(MWh)	(MWh)	(MWh)	(MWh)	(kg/s)	(kg/s)	(kg/s)
0	10442.183	308.507	2.219	0.000	2.219	0.000	0.152	0.152
0.1	10442.671	308.507	2.269	0.080	2.190	0.002	0.150	0.152
0.2	10443.262	308.507	2.330	0.176	2.154	0.004	0.147	0.152
0.3	10443.994	308.507	2.405	0.296	2.110	0.008	0.144	0.152
0.4	10444.925	308.507	2.501	0.448	2.053	0.011	0.140	0.152
0.5	10446.147	308.507	2.627	0.648	1.979	0.016	0.135	0.152
0.6	10447.824	308.507	2.799	0.922	1.877	0.023	0.128	0.152
0.7	10450.266	308.507	3.050	1.321	1.729	0.034	0.118	0.152
0.8	10454.152	308.507	3.450	1.956	1.494	0.050	0.102	0.152
0.9	10461.302	308.507	4.184	3.124	1.061	0.079	0.073	0.152
1	10478.799	308.507	5.983	5.983	0.000	0.152	0.000	0.152

Power Network

Gas Network

$$Q = C_{pipe} \sqrt{\frac{P_i^2 - P_j^2 - E}{GZ}}$$



Research

Heat Decarbonisation North of Tyne (NoT)

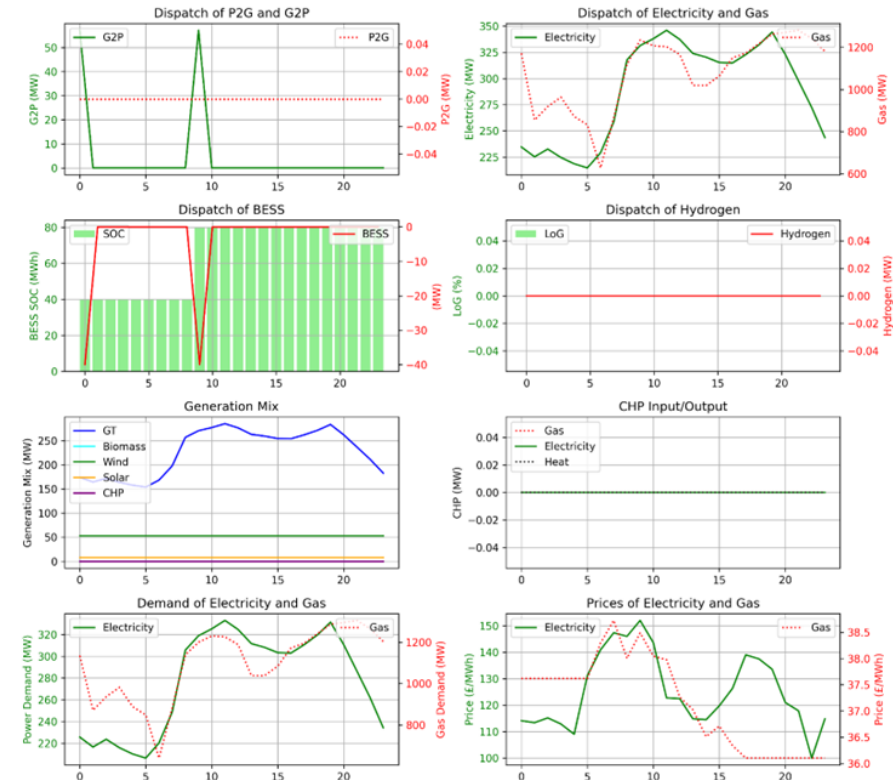


HI ACT: Hydrogen Integration for Accelerated Energy Transitions

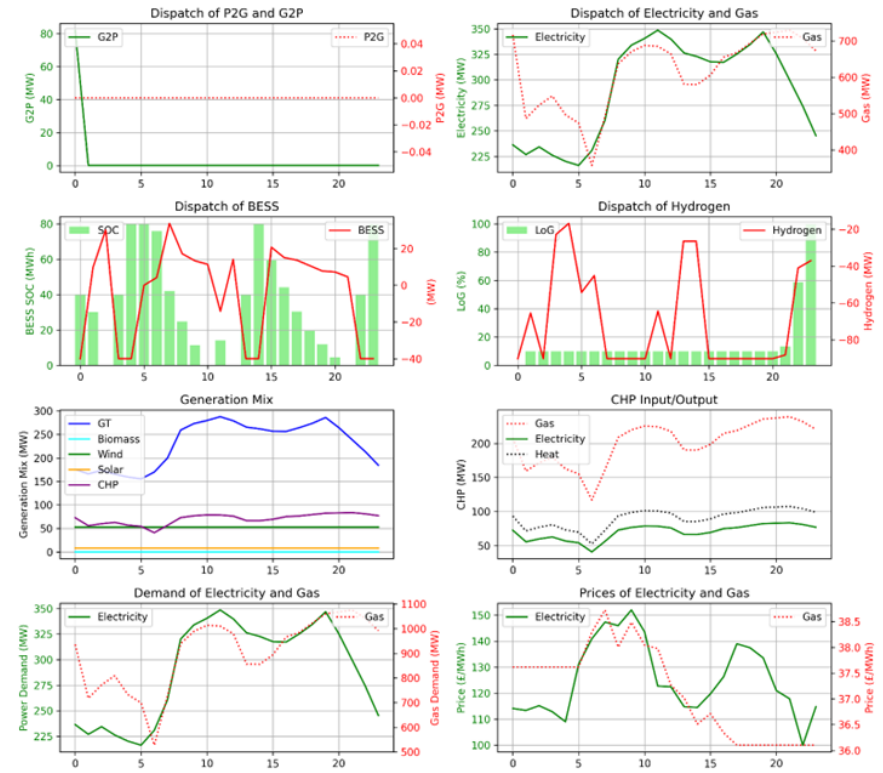
Analysis the economic dispatch for the base case (without interventions)
and Monte Carlo simulation of the social and technical interventions.

BESS: (Positive) Discharging, (Negative) Charging.	Hydrogen: (Positive) Fuel Cell, (Negative) Electrolyser.	Electricity (G2P), Gas (P2G): (Positive) Utilize or Importing, (Negative) Export.
--	--	---

(Base Case)



(Monte Carlo Simulation of Interventions)

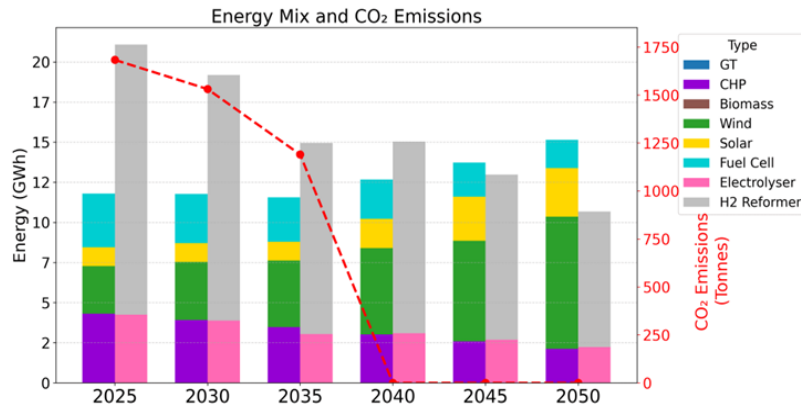


Research

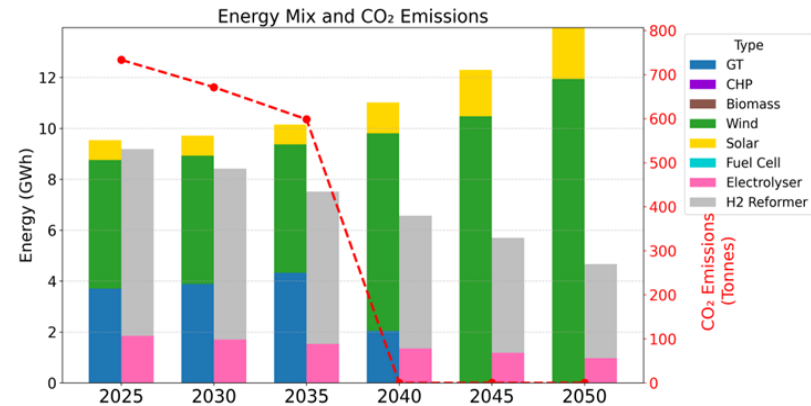
Heat Decarbonisation North of Tyne (NoT)

HI ACT: Hydrogen Integration for Accelerated Energy Transitions

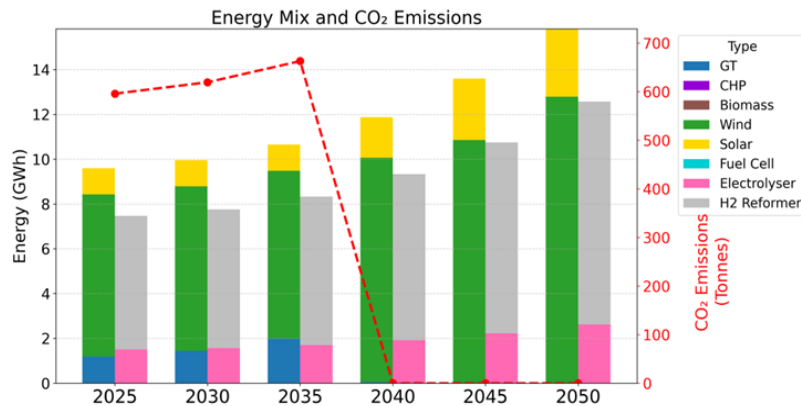
Investment planning (100% hydrogen blending, CCS from 2040)



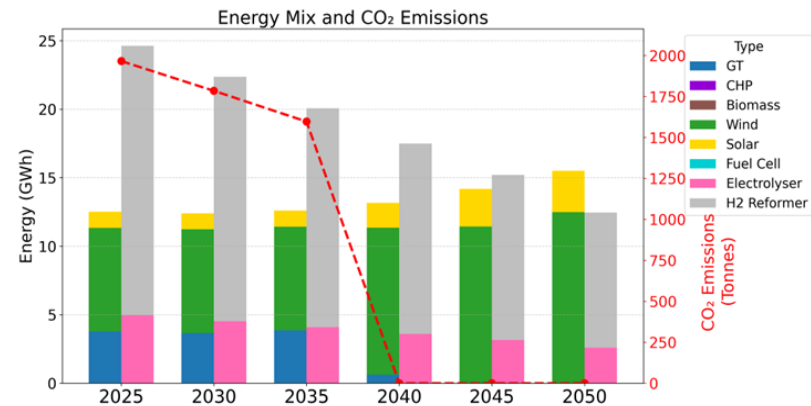
(a) Socio-technological interventions (STI)



(b) 50% heat pumps and 50% hydrogen



(c) 100% heat pumps (100% HP)



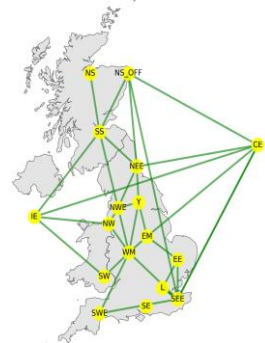
(d) 100% hydrogen boilers (100% HB)

Research

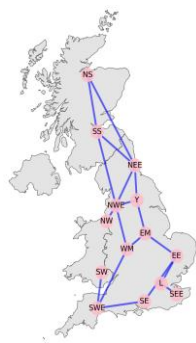
GB's National-Scale Multi-Energy Model

HI ACT: Hydrogen Integration for Accelerated Energy Transitions

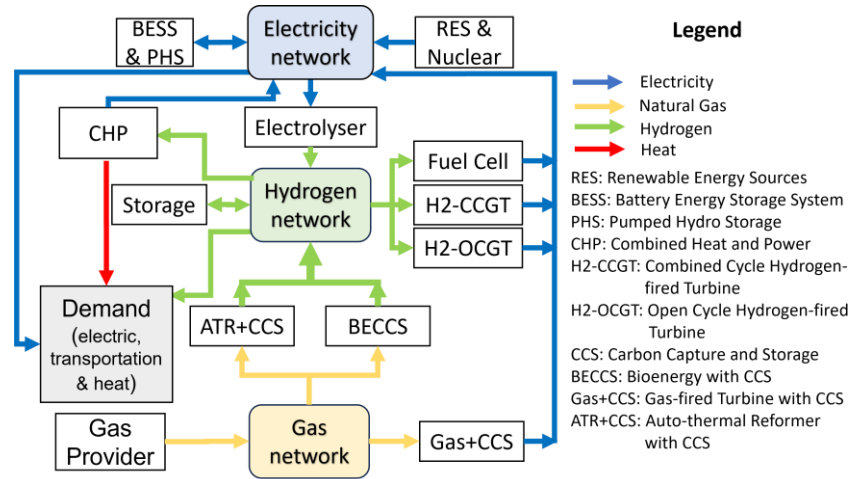
The GB's energy system:
Electricity and gas networks have been built using pandapower and pandapipes in Python.



Electricity Network



Gas/Hydrogen Network

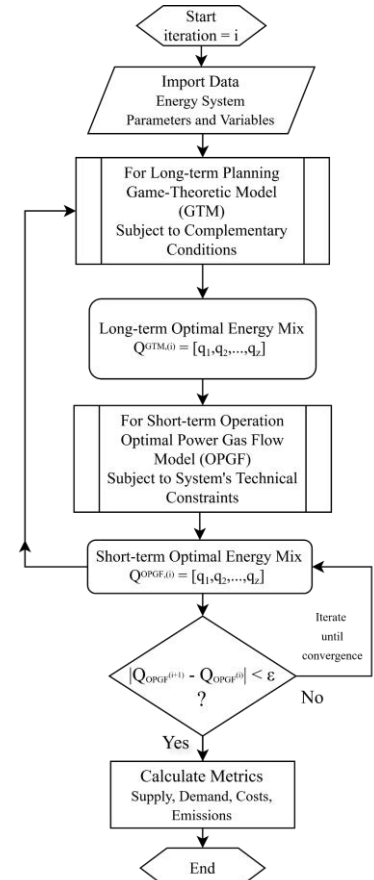
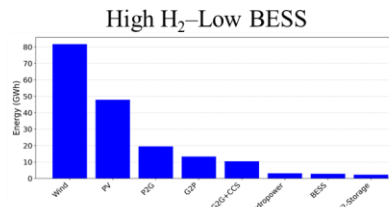
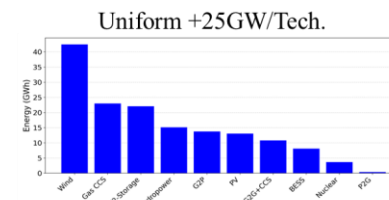
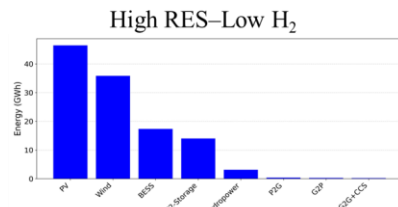
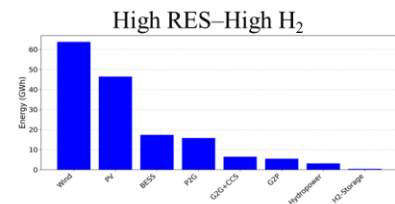


Integrated multi-vector energy system framework

Results:

Simulation results for 2050 peak demand under different capacity allocation scenarios. The electric peak demand is 102.17GWh, while hydrogen peak demand=14.27 GWh.

Metric	Uniform +25GW/Tech.	High RES-High H ₂	High RES-Low H ₂	High H ₂ -Low BESS
Total Demand [GWh]	152.51	158.1	117.75	182.75
Electric Demand [GWh]	119.2	136.22	103.1	149.44
Hydrogen Demand [GWh]	33.31	21.88	14.65	33.31
Total Energy Supply [GWh]	152.51	158.1	117.75	182.75
Total Operational Cost [£]	870,046	862,128	581,663	979,281
Total CO ₂ Emissions [Tonnes]	734	0	0	0



Flowchart for Short-term operation & long-term planning

Research

GB's National-Scale Game Theory Approaches

HI ACT: Hydrogen Integration for Accelerated Energy Transitions

Competitive Planning (Nash-Cournot) vs. Cooperative Planning (Shapley Values)

Each player's decision is driven by maximizing their Net Present Value (NPV).

$$NPV = AF \left(\sum_{t=0}^T (R_t - C_t) - FC \right) - NI$$

where:

$$AF = \frac{(1+r)^n - 1}{r(1+r)^n}$$

r = Discount rate, n

= years of the planning horizon

R_t = Revenue at time t

C_t = Opex cost at time t (£)

$$FC = \sum_{z \in Z} (\phi_z \cdot \varepsilon_z \cdot (K_z + I_z))$$

ϕ_z = Fixed cost fraction of technology z (%)

ε_z = Capex cost of technology z (£/MW)

K_z = Initial capacity of technology z (MW)

I_z = Invested capacity of technology z (MW)

NI = Net investment outlays (£)

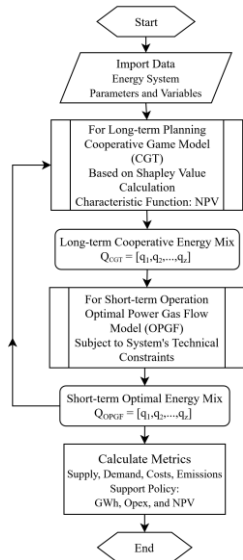
The scenario with higher hydrogen integration demonstrate greater cost reductions and system efficiency under cooperative planning.

2025-2050 **Competitive** Planning (investment span 25 years)

Metric	Scenarios			
	Uniform +25GW/Tech.	High RES–High H ₂	High RES–Low H ₂	High H ₂ –Low BESS
Total Demand [GWh]	182.75	152.06	117.44	223.29
Electric Demand [GWh]	149.44	130.17	102.96	186.18
Hydrogen Demand [GWh]	33.31	21.88	14.48	37.11
Total Energy Supply [GWh]	182.75	152.06	117.44	223.29
Total Operational Cost [m£]	2.22	1.67	1.12	3.09
Total CO ₂ Emissions [Tonnes]	0	0	0	0

2025-2050 **Cooperative** Planning (investment span 25 years)

Metric	Scenarios			
	Uniform +25GW/Tech.	High RES–High H ₂	High RES–Low H ₂	High H ₂ –Low BESS
Total Demand [GWh]	137.81	132.95	116.99	136.86
Electric Demand [GWh]	114.76	114.73	102.5	114.73
Hydrogen Demand [GWh]	23.05	18.23	14.5	22.14
Total Energy Supply [GWh]	137.81	132.95	116.99	136.86
Total Operational Cost [m£]	1.47	1.41	1.14	1.34
Total CO ₂ Emissions [Tonnes]	0	0	0	0



Cooperative Planning

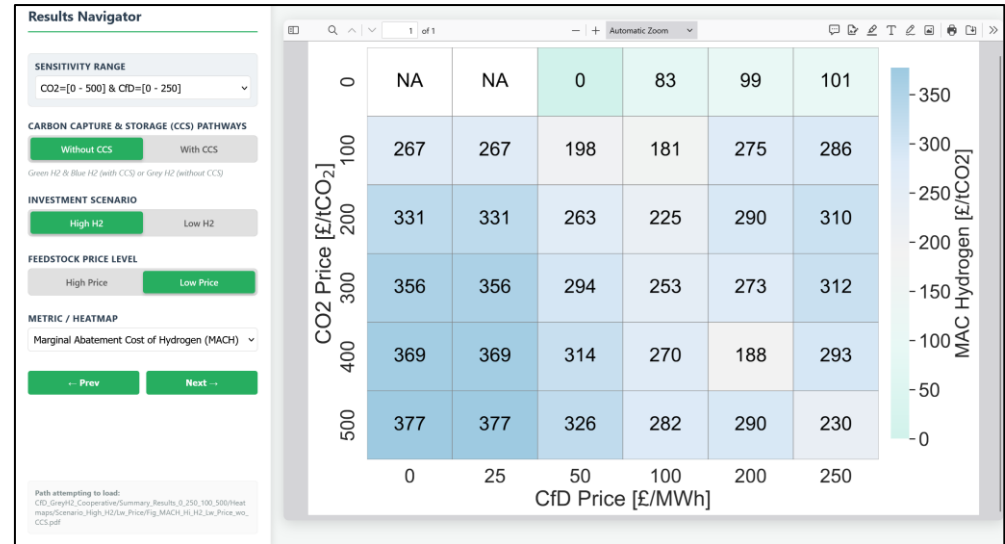
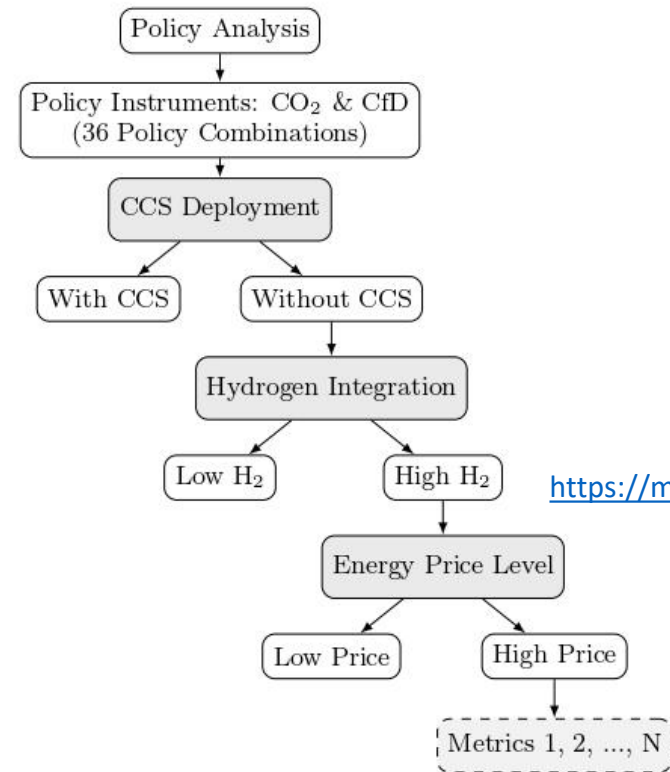
Type	Shapley Value	Norm_Shaps (%)
g2g	1.71E+11	59.19
p2g	1.18E+11	40.81
Onshore Wind	1.02E+11	17.46
Offshore Wind	9.37E+10	16.08
g2p(H2-CCGT)	8.73E+10	14.98
g2p(Fuel Cell)	6.58E+10	11.29
PV	4.29E+10	7.37
Hydro reservoir	4.09E+10	7.02
Storage	3.73E+10	6.4
Nuclear	3.47E+10	5.95
Other RES	2.87E+10	4.93
Hydro ROR	2.71E+10	4.65
g2p(H2-OCGT)	2.25E+10	3.86
Gas CCS	-3.23E+09	1.05
Geothermal	-1.29E+10	4.22
Biomass	-6.47E+10	21.11
Industrial CHP	-8.63E+10	28.15
Micro CHP	-1.39E+11	45.47

Research

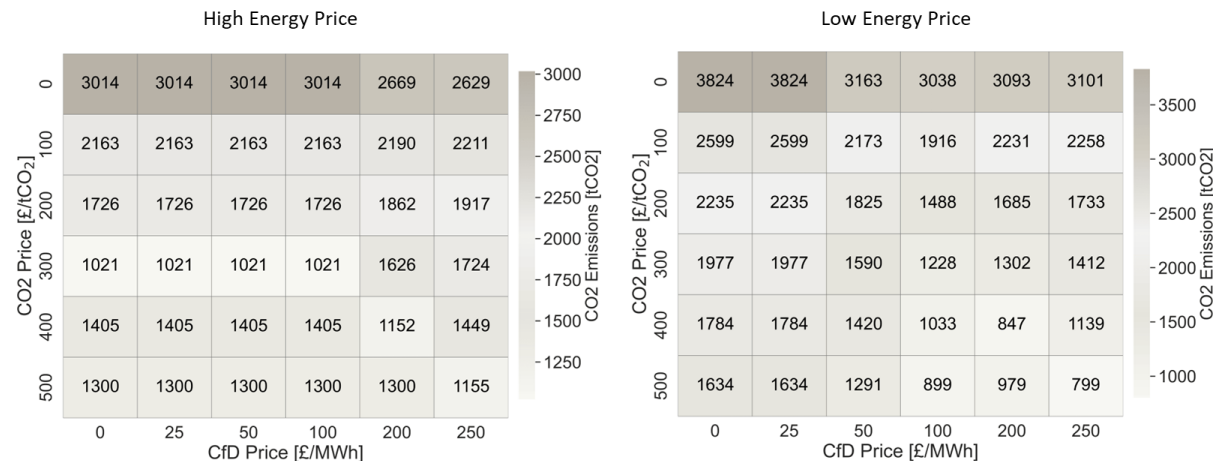
Hydrogen Integration in GB's Energy Policy

HI ACT: Hydrogen Integration for Accelerated Energy Transitions

Graphical Abstract



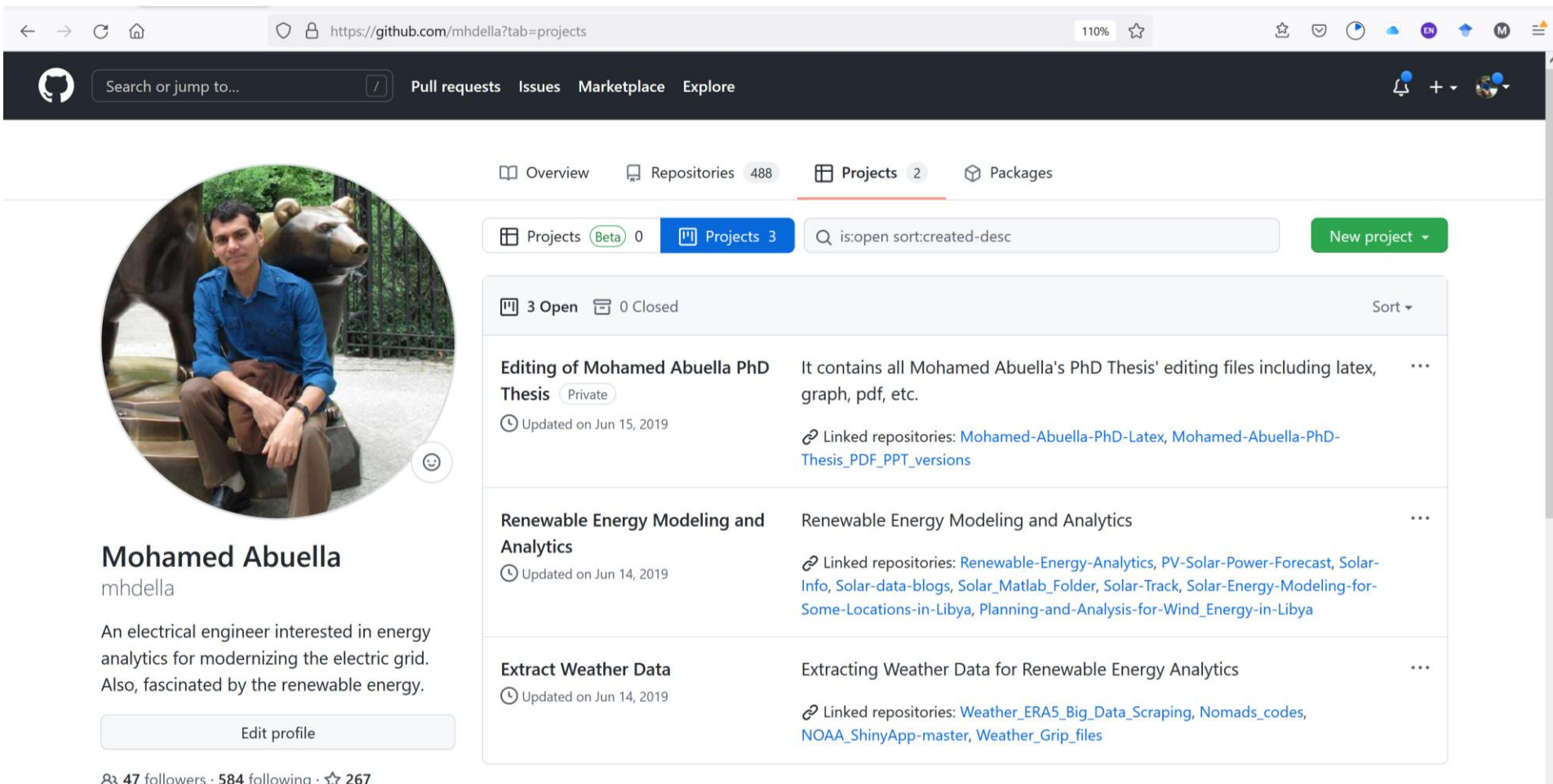
https://mohamedabuella.github.io/HI-ACT/Sim/Interactive_Navigator_Policy_CO2_CfD_Sensitivity.html



CO₂ emissions comparison for high hydrogen integration scenarios without CCS

Research (Miscellaneous)

Some other Projects in GitHub: <https://github.com/mhdella?tab=projects>



The screenshot displays the GitHub profile of Mohamed Abuella (mhdella). The profile includes a circular profile picture of a man sitting on a statue, a bio stating he is an electrical engineer interested in energy analytics, and a list of three open projects. The projects are:

- Editing of Mohamed Abuella PhD Thesis** (Private): It contains all Mohamed Abuella's PhD Thesis' editing files including latex, graph, pdf, etc. Updated on Jun 15, 2019. Linked repositories: Mohamed-Abuella-PhD-Latex, Mohamed-Abuella-PhD-Thesis_PDF_PPT_versions.
- Renewable Energy Modeling and Analytics**: Renewable Energy Modeling and Analytics. Updated on Jun 14, 2019. Linked repositories: Renewable-Energy-Analytics, PV-Solar-Power-Forecast, Solar-Info, Solar-data-blogs, Solar_Matlab_Folder, Solar-Track, Solar-Energy-Modeling-for-Some-Locations-in-Libya, Planning-and-Analysis-for-Wind_Energy-in-Libya.
- Extract Weather Data**: Extracting Weather Data for Renewable Energy Analytics. Updated on Jun 14, 2019. Linked repositories: Weather_ERA5_Big_Data_Scraping, Nomads_codes, NOAA_ShinyApp-master, Weather_Grip_files.

The profile also shows 47 followers, 584 following, and 267 stars.

Research (Miscellaneous)

Keep up some blogs on : <https://mohamedabuella.github.io/blog/>

Blogs

- 22 Mar 2026 » [Blog Tools for Energy Systems Modeling and Analysis](#)
- 17 Dec 2025 » [Blog Interactive Policy Navigator for CO2 and CfD in Energy Systems](#)
- 09 Jun 2025 » [Blog Interactive Levelized Cost of Thermal Energy \(LCOT\) Calculator](#)
- 06 May 2025 » [Blog Decarbonizing Residential Heating in the North of Tyne](#)
- 06 Apr 2025 » [Blog Interactive Tool for Long-Term Planning of Different Energy Assets](#)
- 04 Apr 2025 » [Blog Profit Allocation in a Simple Energy System Using Cooperative Game Theory](#)
- 05 Oct 2024 » [Blog Hydrogen Blending for Sustainable Energy: A Case Study in the North of Tyne Region](#)
- 01 Jun 2024 » [Blog Hydrogen Integration: Lessons from Solar Energy Integration](#)
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- 30 Jul 2019 » [Blog Net Load Forecasting for Microgrid Resiliency](#)
- 29 Jul 2019 » [Blog How a Subtle Lack of Knowledge Could Lead to Catastrophic Consequences](#)
- 23 Jun 2019 » [Blog Reading a Big-picture Book after a While of Focusing on Elaborate Technical Stuff](#)
- 17 Jun 2019 » [website Website Launched](#)

Research (Miscellaneous)

- Power System Flexibility and DG resources management, I have been working on Forecasting and Machine Learning approaches, since 2014
- Techno-economic analysis of HOMER, NREL SAM, and PVLlib Toolbox for Python.
- Writing using Latex (Eqs, Biblio.), Mendeley (~14300 docs) Evernote (organize notes, share them), Dropbox, Google Drive (clouds to back up), iCalendar, etc.
- Research Outreach and Knowledge Dissemination: depending extensively on the online tools, such as Blogs on personal website, LinkedIn, Twitter, Researchgate, Newsletter from relevant groups of interest (ESIG, AI in Smart Grids, ISES, WEMC, etc.)
- Review of IEEE Transactions on Sustainable Energy

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- 27 Receive Recognition on Publons
- Invitations
- Legacy Instructions

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Scores Submitted

Items per page: 10 1 - 10 of 33

ACTION	COMPLETED	ID/TITLE	STATUS
Select...	05-May-2020	TSTE-00996-2019.R2	Under Review

Thanks for Your Listening

Any Question?

Mohamed Ali Abuella
mhdabuella@gmail.com



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UNC CHARLOTTE
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